



Lab 11 (Mini-Project 2)
Fully-Differential Folded Cascode OTA
Design

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Part 1: *gm/ID* Design Charts

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Part 1: g_m/I_D Design Charts

Using ADT Sizing Assistant (Implemented inside Cadence), plot the following design charts vs g_m/I_D for both PMOS and NMOS. Set V_{DS} to a reasonable value and $L = \min, 1\mu, 1\mu, 5\mu$

- V_A
- W/I_D
- f_T
- V_{GS}

For NMOS:

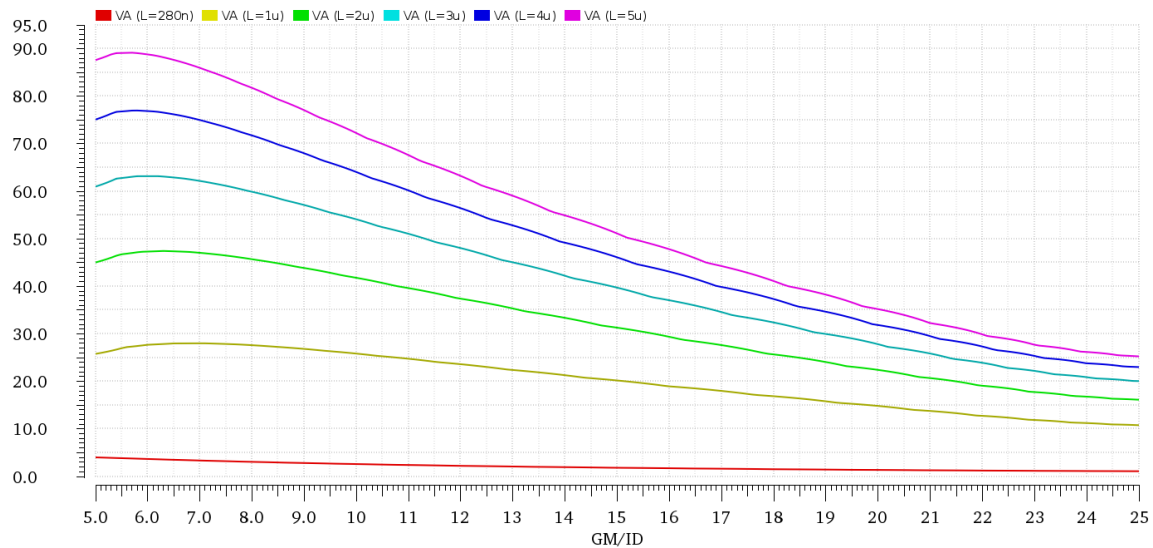


Fig.1: V_A vs g_m/I_D

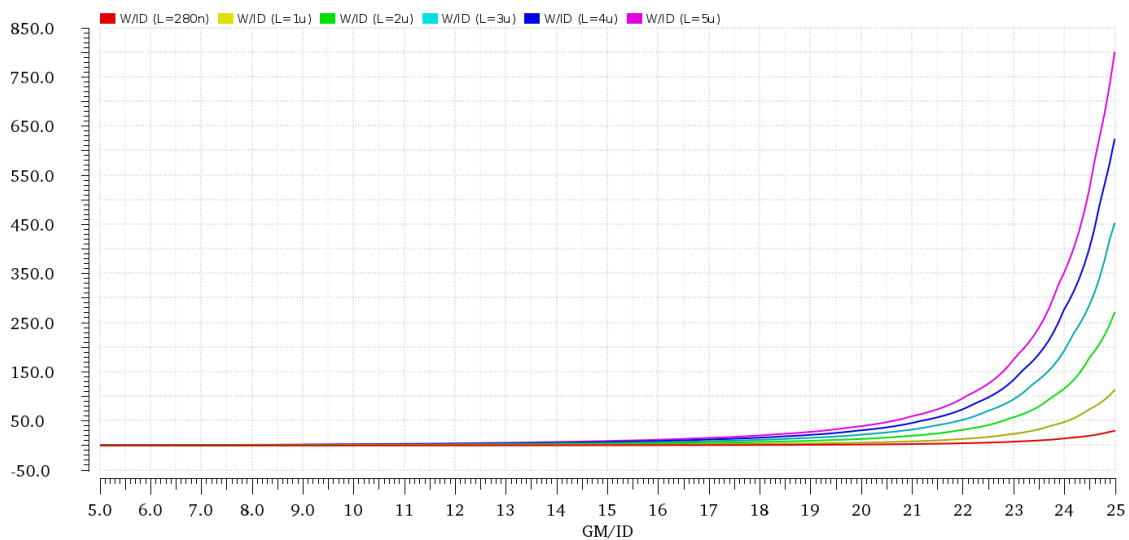


Fig.2: W/I_D vs g_m/I_D

Part 1: g_m/I_D Design Charts

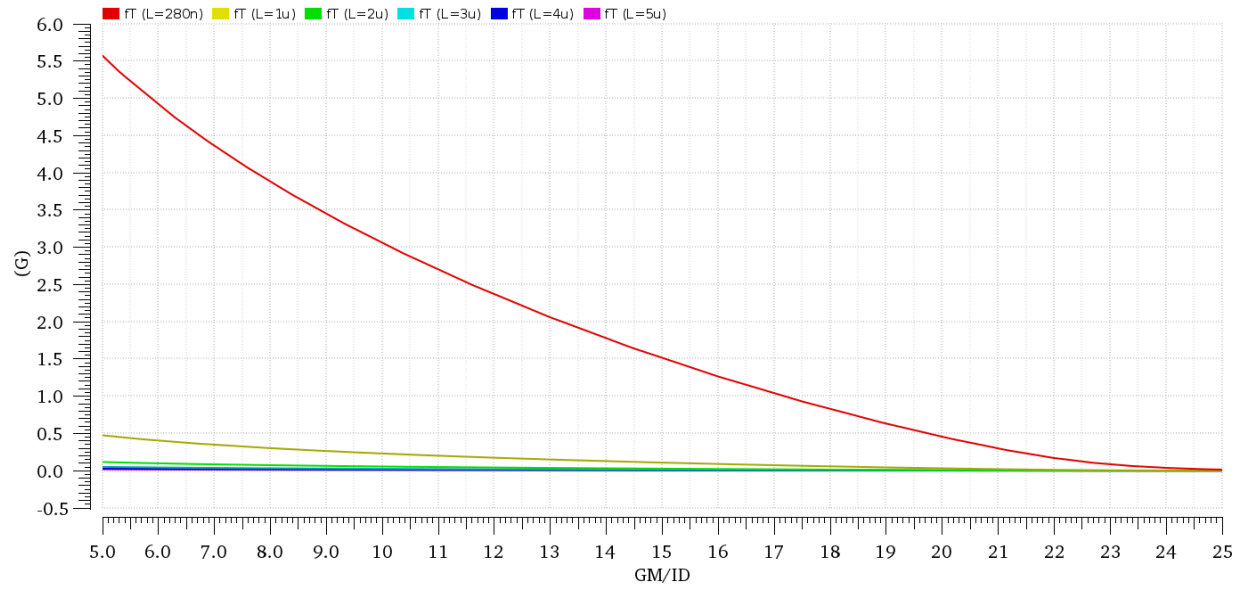


Fig.3: f_T vs g_m/I_D

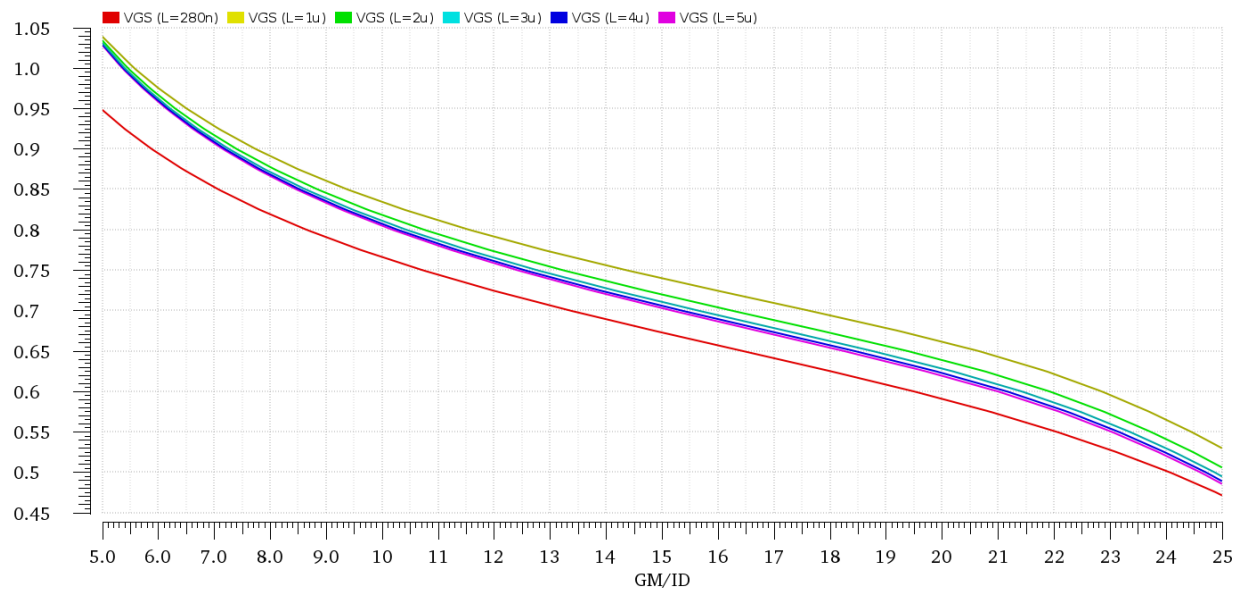


Fig.4: V_{GS} vs g_m/I_D

Part 1: gm/ID Design Charts

For PMOS:

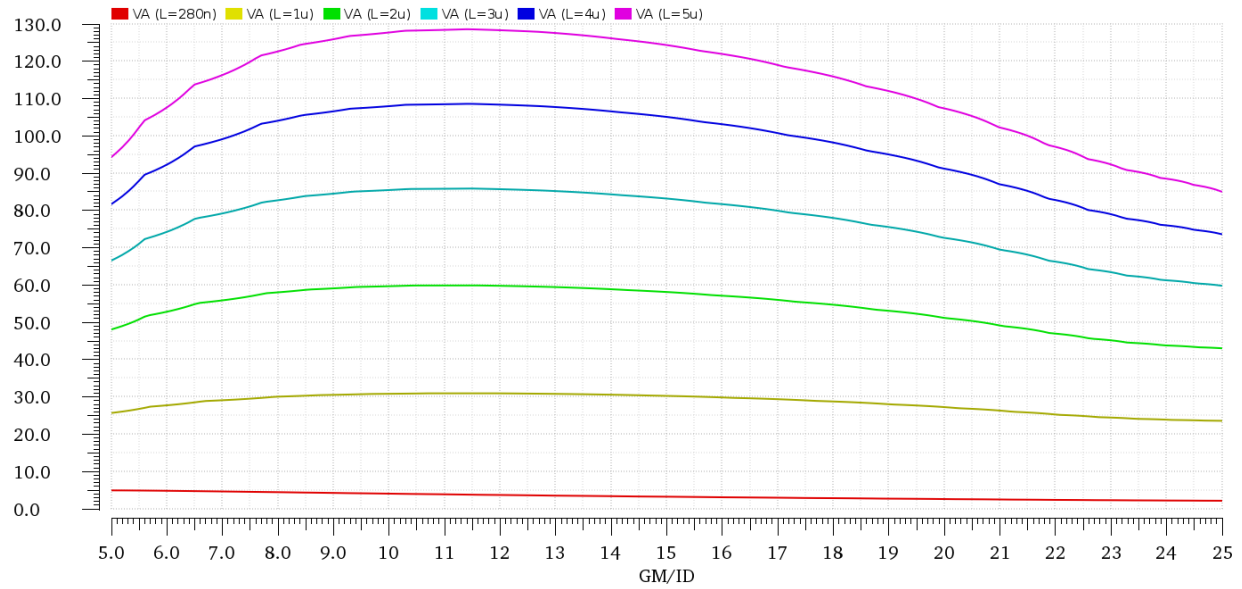


Fig.5: V_A vs g_m/I_D

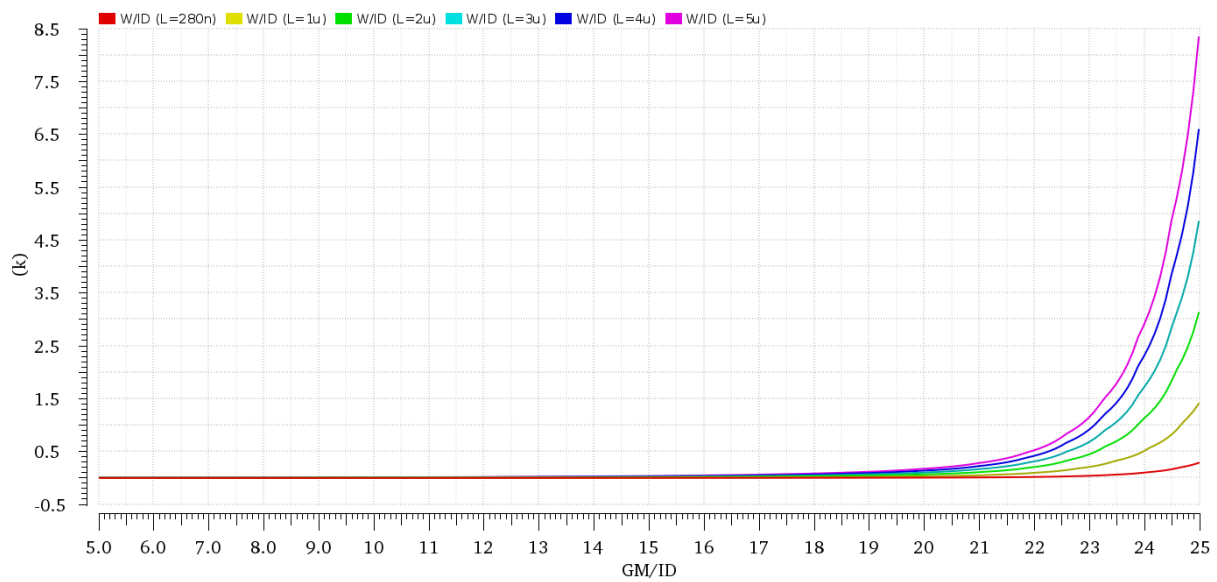


Fig.6: W/I_D vs g_m/I_D

Part 1: g_m/I_D Design Charts

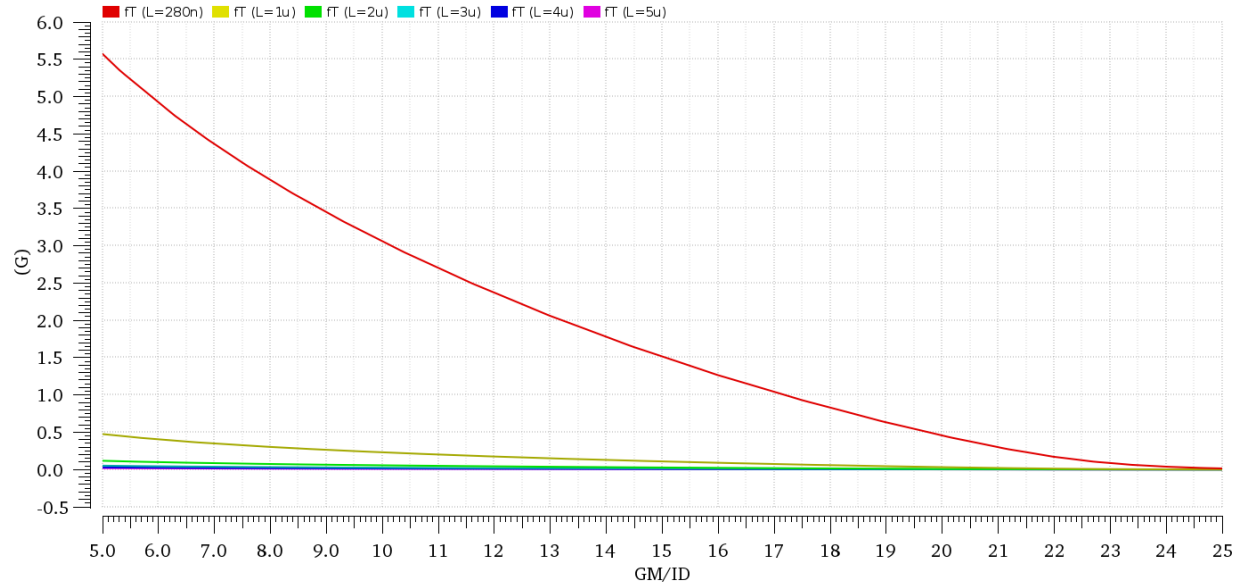


Fig.7: f_T vs g_m/I_D

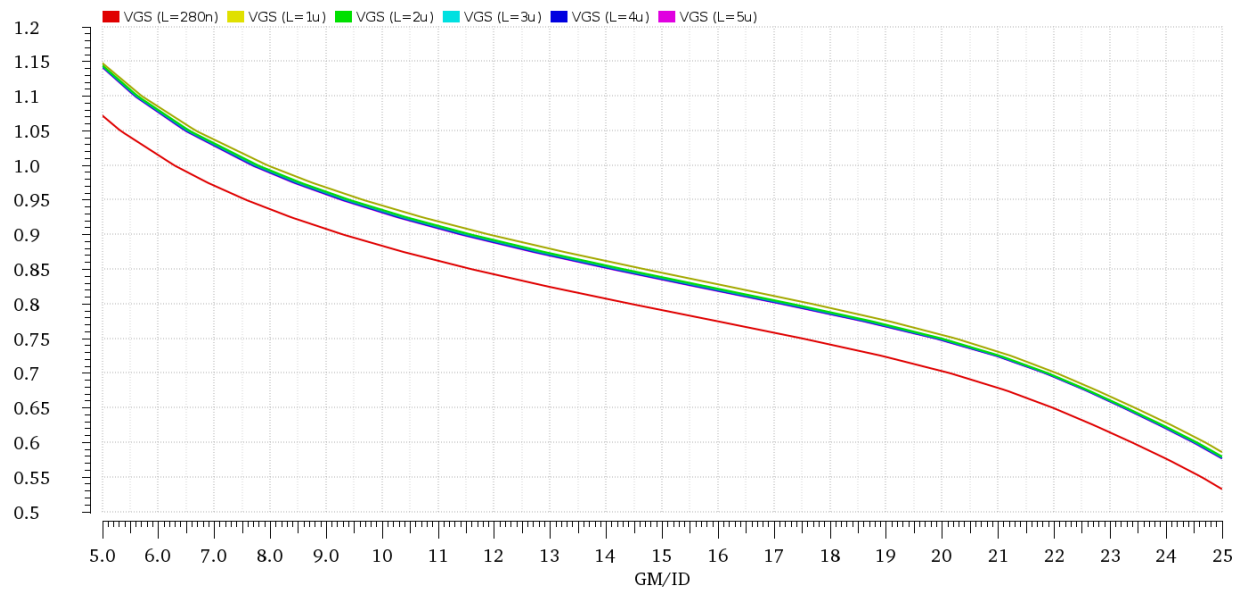


Fig.8: V_{GS} vs g_m/I_D

Part 2: OTA Design

Required Specs.

Design a Fully-Differential Folded Cascode OTA with capacitive feedback that meets the specifications below. The current consumed in the biasing branches (current mirrors) is not included in the specifications.

<i>Parameter</i>	<i>Values</i>
<i>Technology</i>	0.18 μ
<i>Supply Voltage V_{DD}</i>	2.5 V
<i>Closed Loop Gain $A_{v_{CL}}$</i>	2
<i>Phase Margin (in required $A_{v_{CL}}$)</i>	$\geq 70^\circ$
<i>CMIR – low</i>	≤ 0
<i>CMIR – high</i>	≥ 1 V
<i>Differential Output Swing</i>	1.2V <i>peak to peak</i>
<i>Load C_L</i>	500 fF
<i>DC Loop Gain LG</i>	60 dB
<i>Closed Loop settling time t_s</i>	100 ns

Table.1: Required Specs

Note: The power consumption and the total area should be reasonable.

Note: The DC loop gain and closed loop bandwidth specs may be difficult to include (precisely) in the initial hand analysis. After following the suggested design procedure, you may tune the circuit on the simulator (while taking trade-offs into account) to achieve these specs.

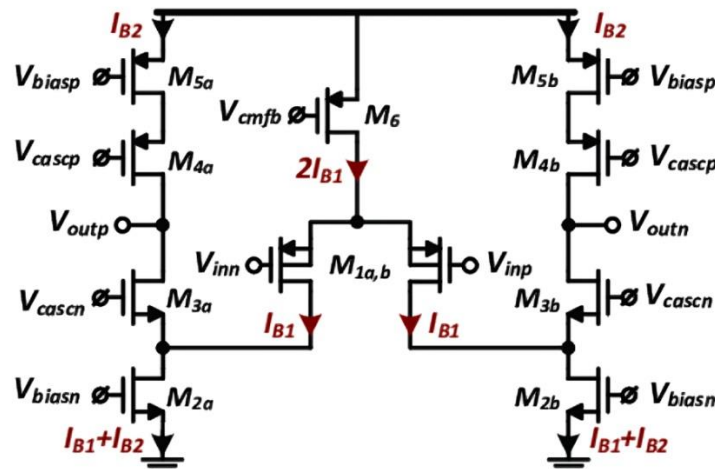


Fig.9: Schematic of fully-differential folded Cascode OTA

Topology Selection

- From required spec on CMIR

We notice it's close to ground, therefore the topology we will use in the Fully-Differential Folded Cascode OTA is PMOS for the CS stage and NMOS for the CG stage.

Pre-Sizing

- From CL testbench we calculate β

$$\beta = \frac{C_f}{C_f + C_{in}} = \frac{1}{3}$$

$$LG = \beta A_{v_{OL}} = 1000$$

$$A_{v_{OL}} = 3000$$

$$A_{v_{OL}} = 69.55 \text{ dB}$$

- From Spec. of CL Settling time for 1% error = 100n

$$V_{out} = V_{in} \left(1 - e^{-\frac{t}{\tau}} \right)$$

$$e^{-\frac{t}{\tau}} = 0.01$$

$$\ln(0.01) = -\frac{t}{\tau}$$

$$\tau = 21.71 \text{ ns}$$

$$BW_{CL} = \frac{1}{2\pi\tau} = 7.33 \text{ MHz}$$

$$BW_{OL} = \frac{BW_{CL}}{1 + LG} = 7.33 \text{ KHz}$$

$$GBW = A_{v_{OL}} \times BW_{OL} = 22 \text{ MHz}$$

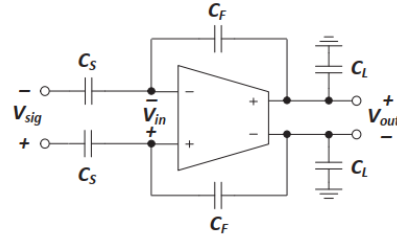
$$C_{Leff} = C_L + (C_f // C_{in})$$

$$GBW = \frac{g_m}{2\pi C_{Leff}}$$

$$g_m = 161.26 \mu\text{S}$$

- We will approximate this g_m with 10 % margin

$$g_m = 175 \mu\text{S}$$



Part 2: OTA Design

- The total bias current of the Folded Cascode is given by

$$I_{B_{total}} = 2I_{B1} + 2I_{B2}$$

Then we define a split ratio (S) as the ratio between the CS and CG bias currents.

$$S = \frac{I_{B1}}{I_{B2}}$$

So, the design requires a settling time $t_s = 100ns$, a $LG = 60dB$ (which is relatively high), and this Figure of Merit as high as possible

$$FOM = \frac{GBW}{Area \times I_{B_{total}}}$$

Based on these specs a splitting ratio (S) with 2 to 2.5 is recommended to maximize GBW and keeping a settling time with $100ns$ and increasing (S) increases the width of the input pair, while decreasing the widths of the output branch transistors.

So, I choose a splitting ratio (S) = 2.24 which means $I_{B1} = 11 \mu A$, and $I_{B2} = 5 \mu A$ to achieve these specs.

Folded Cascode OTA Devices Sizing

<i>Parameters</i>	$M_{1,2}$ <i>Input Pair</i>	M_0 <i>Tail CS</i>	$M_{3,4}$ <i>PMOS CS</i>	$M_{9,10}$ <i>NMOS CS</i>	$M_{5,6}$ <i>NMOS Cascode</i>	$M_{7,8}$ <i>PMOS Cascode</i>
W	30 μm	11 μm	2.56 μm	5.22 μm	840 nm	2.16 μm
L	420 nm	500 nm	500 nm	1.3 μm	320 nm	290 nm
I_D	11 μA	22 μA	5 μA	16 μA	5 μA	5 μA
g_m	175.8 μS	175.3 μS	39.87 μS	127 μS	49.61 μS	49.22 μS
g_m/I_D	16 S/A	8 S/A	8 S/A	8 S/A	10 S/A	10 S/A
$V_{D_{\text{sat}}}$	99.51 mV	215.1 mV	214.9 mV	212.5 mV	158.6 mV	183.1 mV
V_{ov}	45.34 mV	212 mV	211.8 mV	208.4 mV	123.8 mV	153.4 mV

Table.2: Folded Cascode OTA Sizing

<i>Parameters</i>	M_{11} <i>PMOS CM</i>	M_{12} <i>PMOS CB</i>	M_{13} <i>NMOS CM</i>	M_{14} <i>PMOS CB</i>	M_{18} <i>NMOS CB</i>	M_{16} <i>NMOS Magic Battery</i>	M_{19} <i>PMOS Magic Battery</i>
W	$5\ \mu m$	$970\ nm$	$711\ nm$	$540\ nm$	$350\ nm$	$280\ nm$	$270nm$
L	$500\ nm$	$500\ nm$	$1.3\ \mu m$	$500\ nm$	$1.3\ \mu m$	$6\ \mu m$	$2.5\ \mu m$
I_D	$10\ \mu A$	$2\ \mu A$	$2\ \mu A$	$1.2\ \mu A$	$1.2\ \mu A$	$1.2\ \mu A$	$1.2\ \mu A$
g_m	$79.68\ \mu S$	$16\ \mu S$	$15.94\ \mu S$	$9.53\ \mu S$	$9.45\ \mu S$	$3.514\ \mu S$	$2.42\ \mu S$
g_m/I_D	$8\ S/A$	$8\ S/A$	$8\ S/A$	$8\ S/A$	$8\ S/A$	$2.5\ S/A$	$2.23\ S/A$
V_{Dsat}	$215.1\ mV$	$214.4\ mV$	$211.4\ mV$	$215.9\ mV$	$214\ mV$	$502.2\ mV$	$668.6\ mV$
V_{ov}	$212.1\ mV$	$211.2\ mV$	$206.9\ mV$	$213.1\ mV$	$210.3\ mV$	$649.4\ mV$	$916\ mV$

Table.3: mirroring branches and magic battery sizing

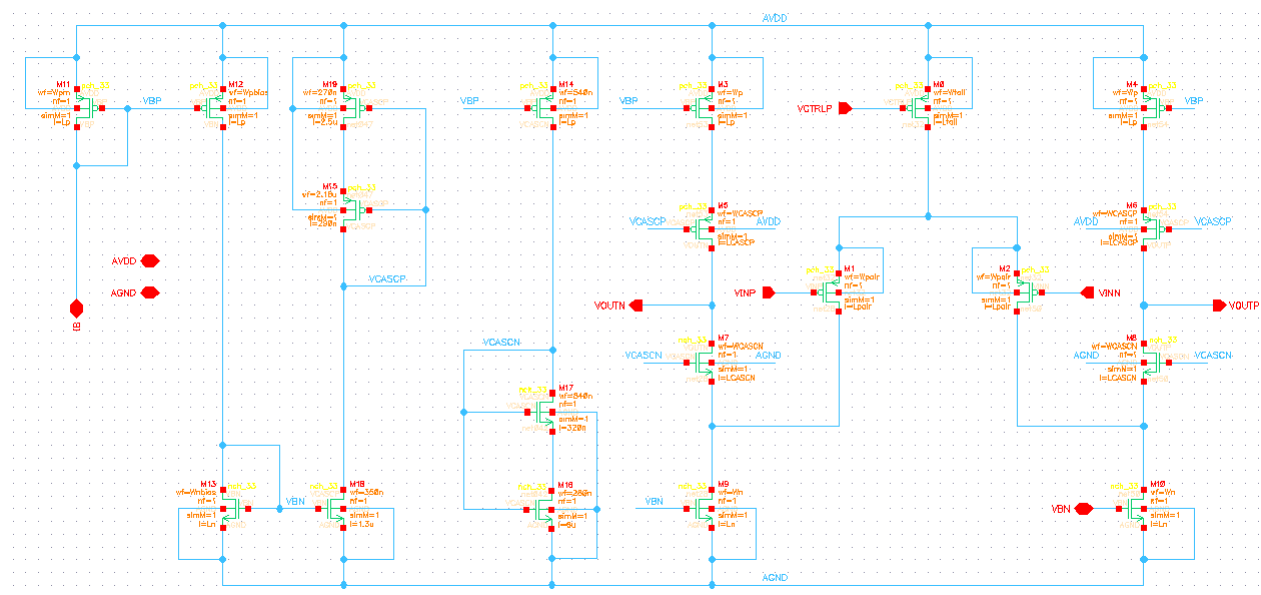


Fig.21: Folded Cascode OTA schematic

Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

We will start with a behavioral CMFB network similar to the one shown below. We use ideal buffers to avoid loading the OTA output with the CM sensing resistors. Note that we don't need high gain in the CMFB loop thus, we use a gain = 1 in the error amplifier. We use dummy pins in the behavioral CMFB circuit to be "pin-accurate" with the actual CMFB circuit we will use later.

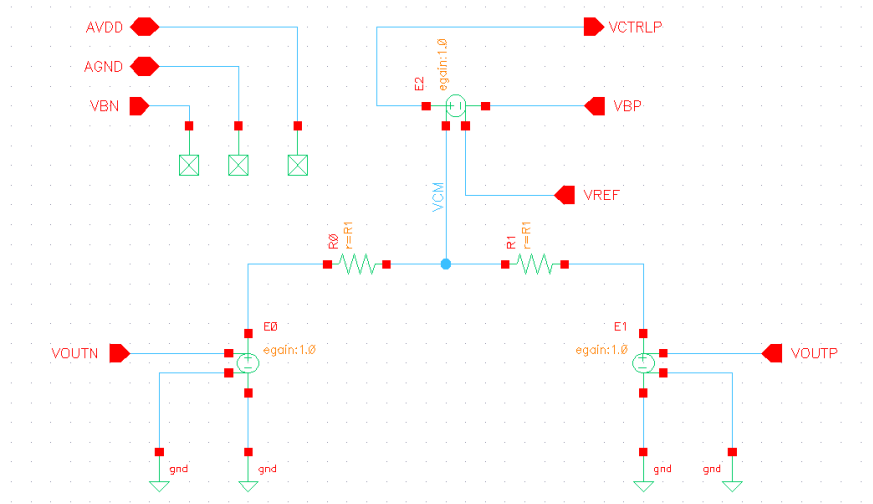


Fig.22: Behavioral CMFB schematic

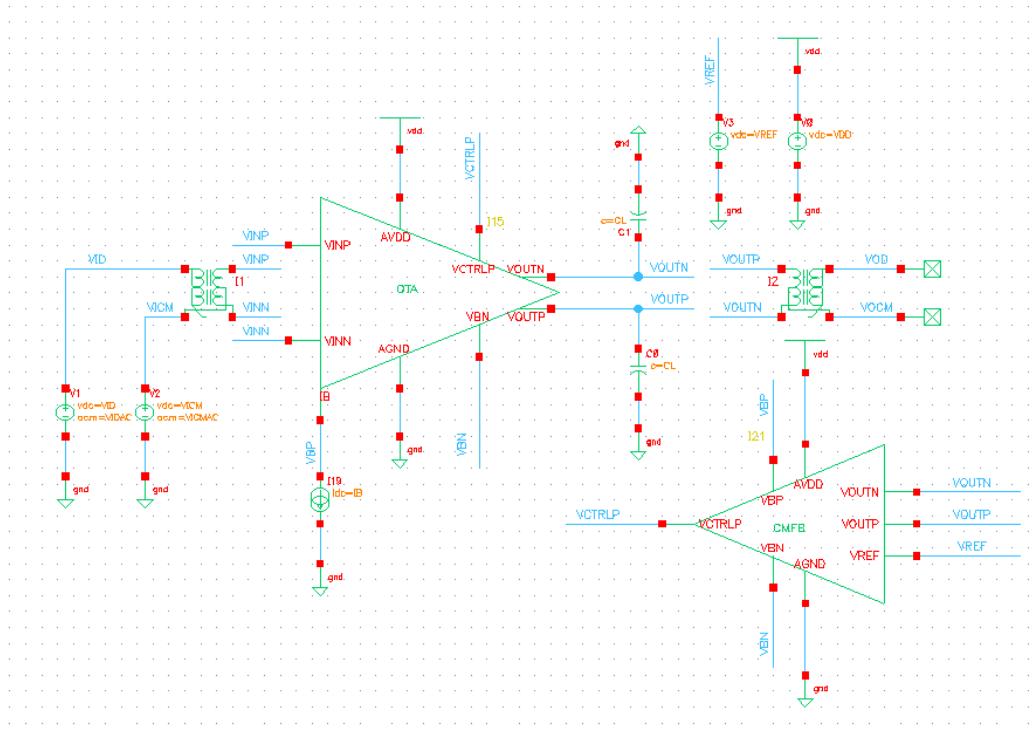
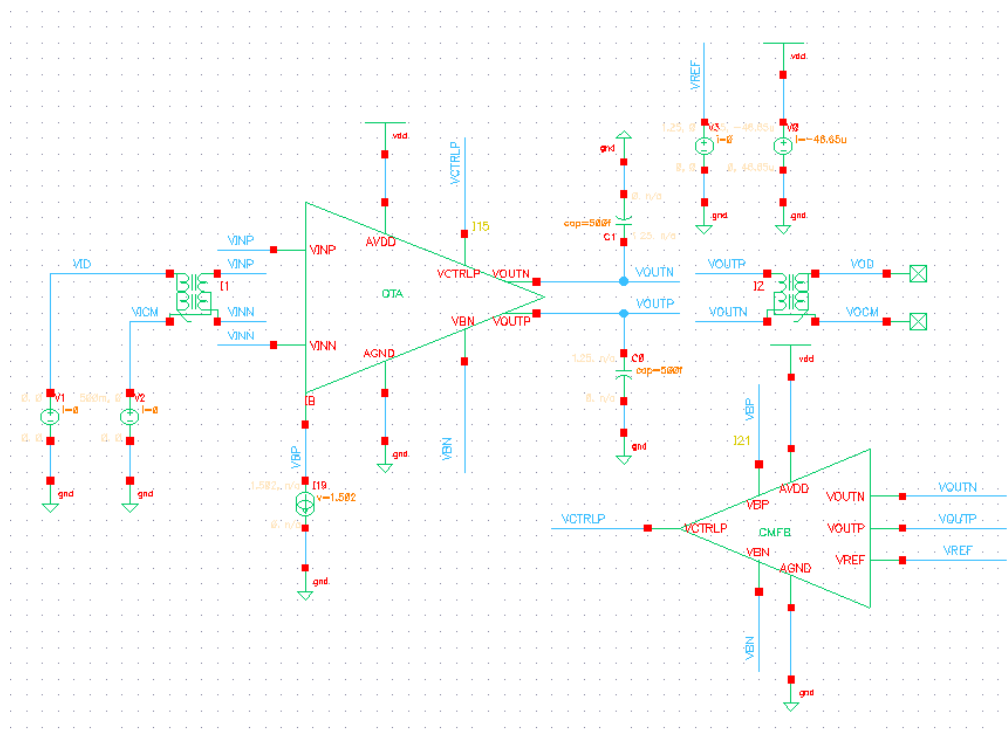


Fig.23: Open Loop testbench



The screenshot shows a circuit simulation environment with two component windows, U1 and U2, overlaid on a schematic diagram. The schematic includes various components like resistors, capacitors, and signal sources, connected by wires. The component windows display a table of parameters and their corresponding values.

Parameters	Value
getData("I15.M1.m0.id"?result "dcOp")	-11.06u
getData("I15.M1.m0.vgs"?result "dcOp")	-827.6m
getData("I15.M1.m0.vds"?result "dcOp")	-891.5m
getData("I15.M1.m0.vdsat"?result "dcOp")	-99m
getData("I15.M1.m0.vth"?result "dcOp")	-782.4m
getData("I15.M1.m0.gm"?result "dcOp")	177u
getData("I15.M1.m0.gmb"?result "dcOp")	74.55u
getData("I15.M1.m0.gds"?result "dcOp")	1.059u
getData("I15.M1.m0.region"?result "dcOp")	2
getData("I15.M1.m0.gmoverid"?result "dcOp")	16.01

Parameters	Value
getData("I15.M2.m0.id"?result "dcOp")	-11.06u
getData("I15.M2.m0.vgs"?result "dcOp")	-827.6m
getData("I15.M2.m0.vds"?result "dcOp")	-891.5m
getData("I15.M2.m0.vdsat"?result "dcOp")	-99m
getData("I15.M2.m0.vth"?result "dcOp")	-782.4m
getData("I15.M2.m0.gm"?result "dcOp")	177u
getData("I15.M2.m0.gmb"?result "dcOp")	74.55u
getData("I15.M2.m0.gds"?result "dcOp")	1.059u
getData("I15.M2.m0.region"?result "dcOp")	2
getData("I15.M2.m0.gmoverid"?result "dcOp")	16.01

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Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

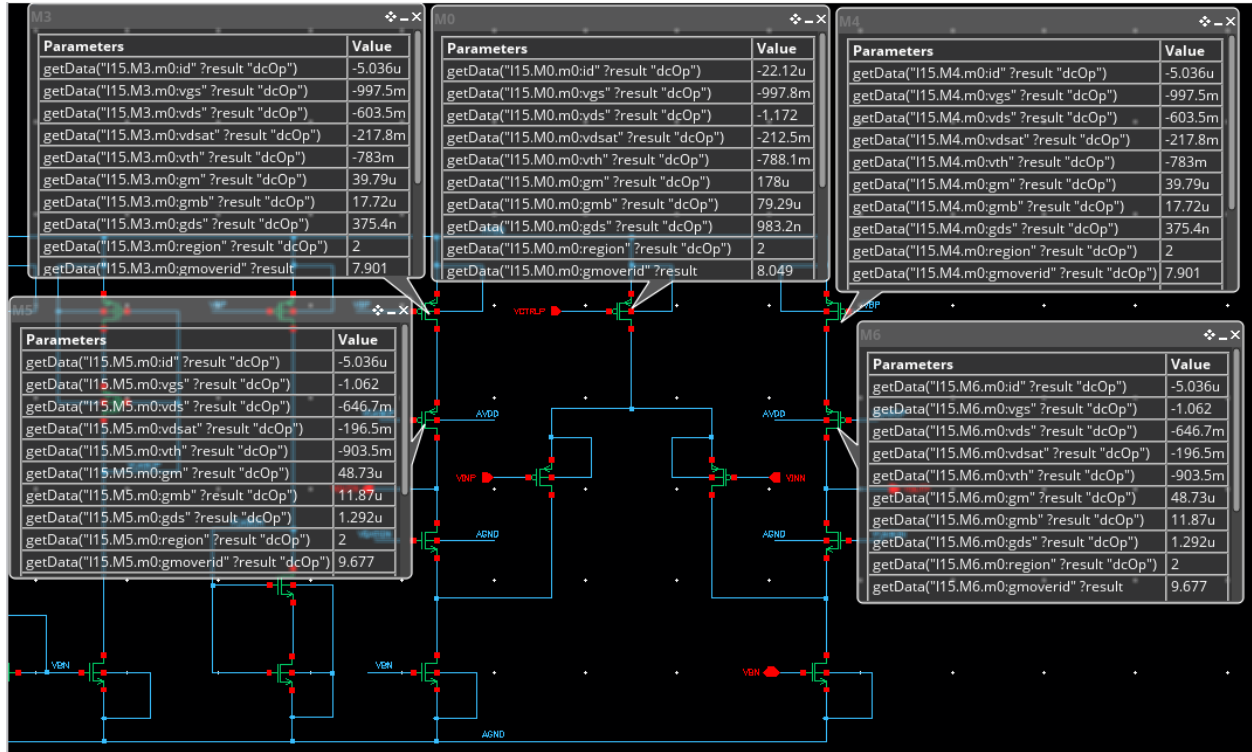


Fig.26: OP of PMOS CSs and PMOS Cascodes



Fig.27: OP of NMOS CSs and NMOS Cascodes

Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

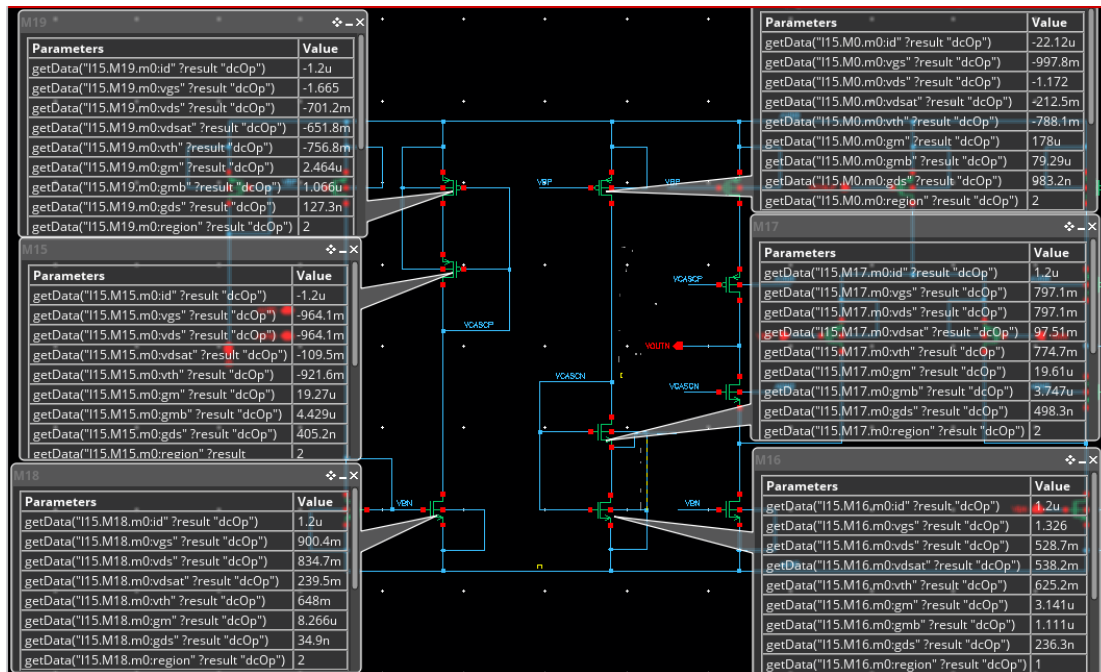


Fig.28: OP of NMOS and PMOS Magic battery

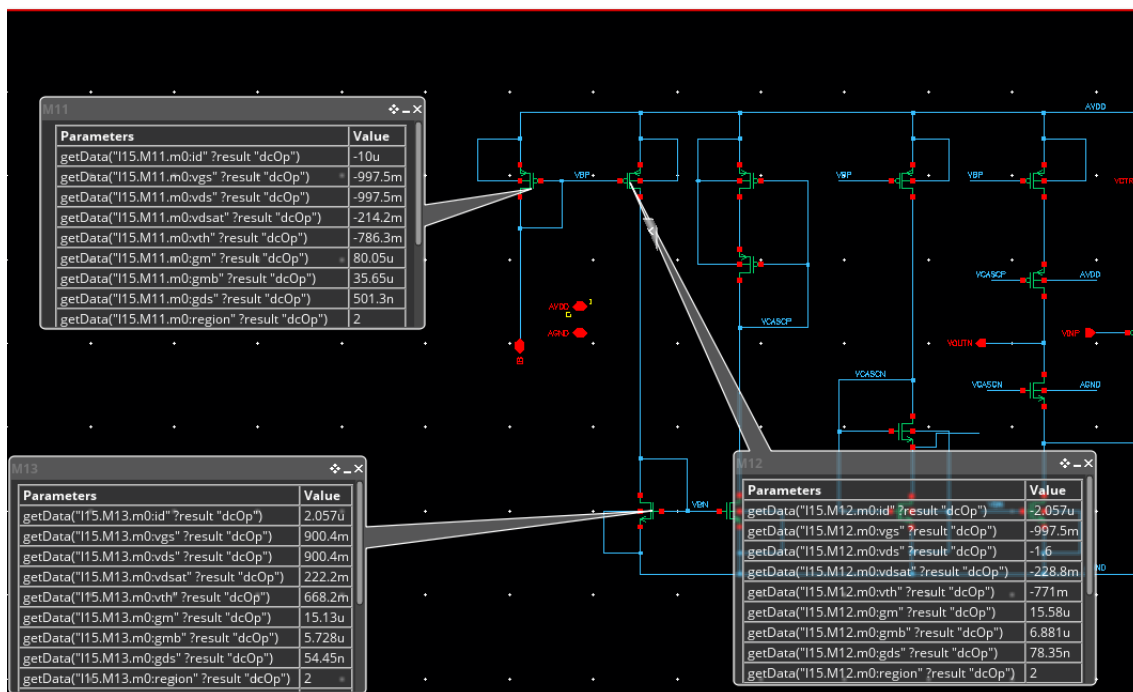


Fig.29: OP of NMOS and PMOS CM

- Note that we don't need high gain in the CMFB loop Why?

Because small error is acceptable.

Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

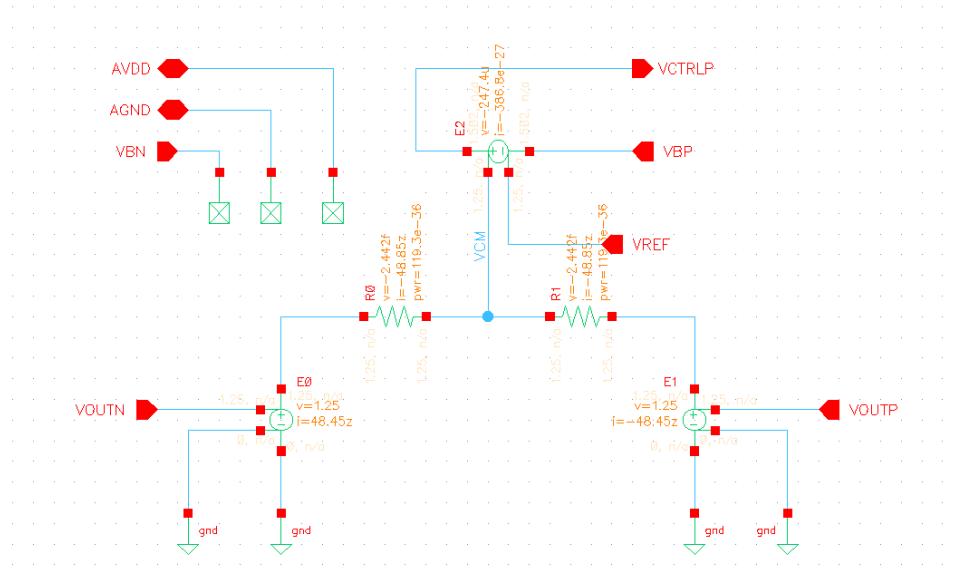


Fig.30: OP of Behavioral CMFB

- All Devices are in saturation except non-degenerated devices of the magic batteries.
- The Output Swing is from $0.45mV$ to $2.04V$, so I choose a $V_{REF} = 1.25V$.
- The total Current Consumption in the Folded Cascode OTA and in current mirror branches.

$$I_{B_{total}} = 46.59 \mu A, \text{ (a } 32.19 \mu A \text{ in the Folded Cascode OTA)}$$

- The dc level of common-mode output $V_{out_{CM}} = V_{REF} = 1.25V$.
- What is the differential input and output voltages of the error amplifier? What is the relation between them?

The differential input and output voltages of the error amplifier are $-250 \mu V$ the same as the relation between them is the error amplifier gain which is 1.

Diff. Small Signal Ccs (AC Analysis)

Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM)

Output	Nominal	Spec	Weight	Pass/Fail
mag(v("/VOD" ?result "ac"))				
dB20(mag(v("/VOD" ?result "ac"))				
phaseDegUnwrapped(v("/VOD" ?result "ac"))				
Ao	3.814k	> 3k		pass
Ao_dB	71.63	> 70		pass
BW	14.29k	> 7.33k		pass
(Ao * BW)	54.52M	> 22M		pass
GBW	54.64M	> 22M		pass
UGF	54.57M	> 22M		pass
Cdd1+Css7+Cdd9	6.358f			
PM	79.8			
VOCM dc	1.25			
VREF dc	1.25			

Fig.31: diff. SS results

Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

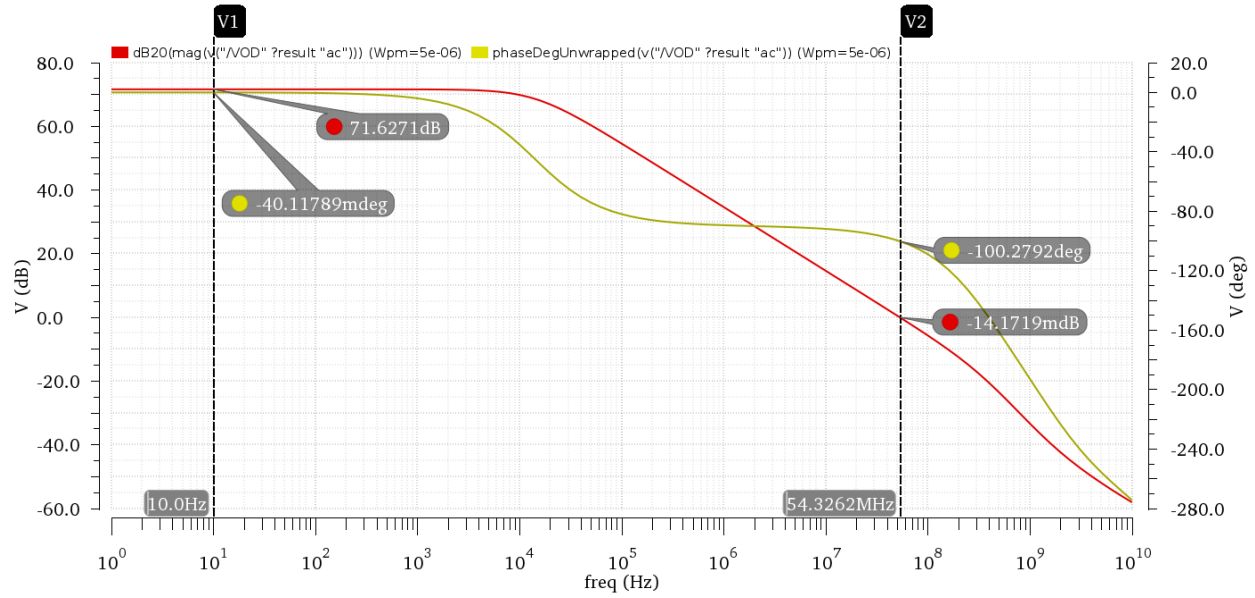


Fig.32: Bode Plot of diff. Ccs

- Hand Analysis (Note that parasitic caps are neglected cause they too small)

$$A_{vd} = G_m \times R_{out} = g_{m1,2} \times (r_{o7}(g_{m7} + g_{mb7})(r_{o9}/r_{o1})/r_{o5}(g_{m5} + g_{mb5})r_{o3})$$

$$\omega_{pd} = \frac{1}{R_{out}C_L}, \quad BW = \frac{\omega_{pd}}{2\pi}$$

$$\omega_{pnd} = \frac{g_{m7}}{C_{dd1} + C_{ss7} + C_{dd9}}$$

$$GBW = A_{vd} \times BW = \frac{g_{m1,2}}{2\pi C_L}$$

$$PM = 90^\circ - \tan^{-1}\left(\frac{\omega_u}{\omega_{pnd}}\right)$$

Parameters	Analytical	Simulation
DC Gain A_{vd}	3.742K	3.814K
DC Gain in dB A_{vd}	71.46 dB	71.63 dB
BW	15.05 KHz	14.29 KHz
GBW	56.34 MHz	54.64 MHz
UGF	54.34 MHz	54.57 MHz
Phase Margin PM	87.5°	79.8°

Table4: diff. small signal comparison

GBW and UGF are almost the same, so that's a good indicator for phase margin PM.

Resistive Sensing Network

The signal swing is

$$V_{out_{max}} = V_{DD} - V_5^* - |V_{GS8}| = 1.4038 V$$

$$V_{out_{min}} = V_8^* + V_0^* = 0.38 \text{ V}$$

$$V_{REF} = V_{out_{CM}} = \frac{V_{out_{max}} + V_{out_{min}}}{2} = 0.9V$$

$$V_{REF} \approx 1V$$

$$R_{min} = \frac{V_{p2p_{min}}}{2 \times I_D} = \frac{1.2}{2 \times 2.5\mu} = 240K\Omega$$

CMFB Devices Sizing

<i>Parameters</i>	$M_{1,2}$	M_0	$M_{3,4}$	<i>PMOS Input devices</i>		$M_{5,6}$	$M_{7,8}$
	<i>Input Pair</i>	<i>Tail CS</i>	<i>PMOS CS</i>	M_9	M_{10}	<i>PMOS CS</i>	<i>PMOS CS</i>
W	$1.44\ \mu m$	$711\ nm$	$520\ nm$	$2.56\ \mu m$	$32\ \mu m$	$2.56\ \mu m$	$32\ \mu m$
L	$1\ \mu m$	$1.3\ \mu m$	$500\ nm$	$500\ nm$	$1\ \mu m$	$500\ nm$	$1\ \mu m$
I_D	$1\ \mu A$	$2\ \mu A$	$1\ \mu A$	$5\ \mu A$	$5\ \mu A$	$5\ \mu A$	$5\ \mu A$
g_m	$14.73\ \mu S$	$15.92\ \mu S$	$8.02\ \mu S$	$39.87\ \mu S$	$74.23\ \mu S$	$39.87\ \mu S$	$74.23\ \mu S$
g_m/I_D	$15\ S/A$	$8\ S/A$	$8\ S/A$	$8\ S/A$	$15\ S/A$	$8\ S/A$	$15\ S/A$
V_{Dsat}	$109.3\ mV$	$211.9\ mV$	$213.5\ mV$	$214.9\ mV$	$95.7\ mV$	$214.9\ mV$	$95.7\ mV$
V_{ov}	$55.7\ mV$	$207.5\ mV$	$210\ mV$	$211.8\ mV$	$63\ mV$	$211.8\ mV$	$63\ mV$

Table.5: CMFB Devices Sizing

Part 4: Open-Loop OTA Simulation (Actual CMFB)

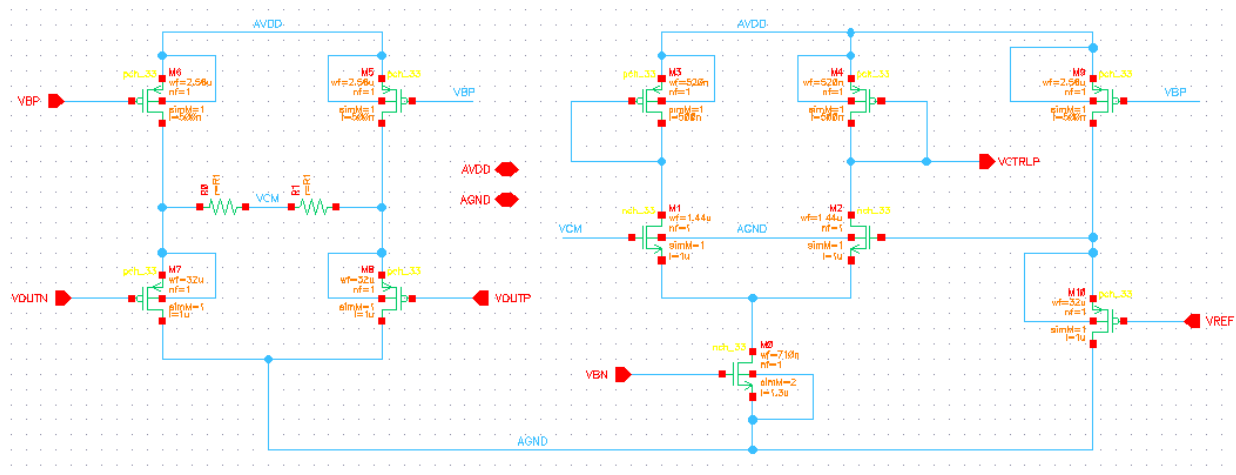


Fig.41: Final Sizing of CMFB schematic

OP and Biasing Voltages

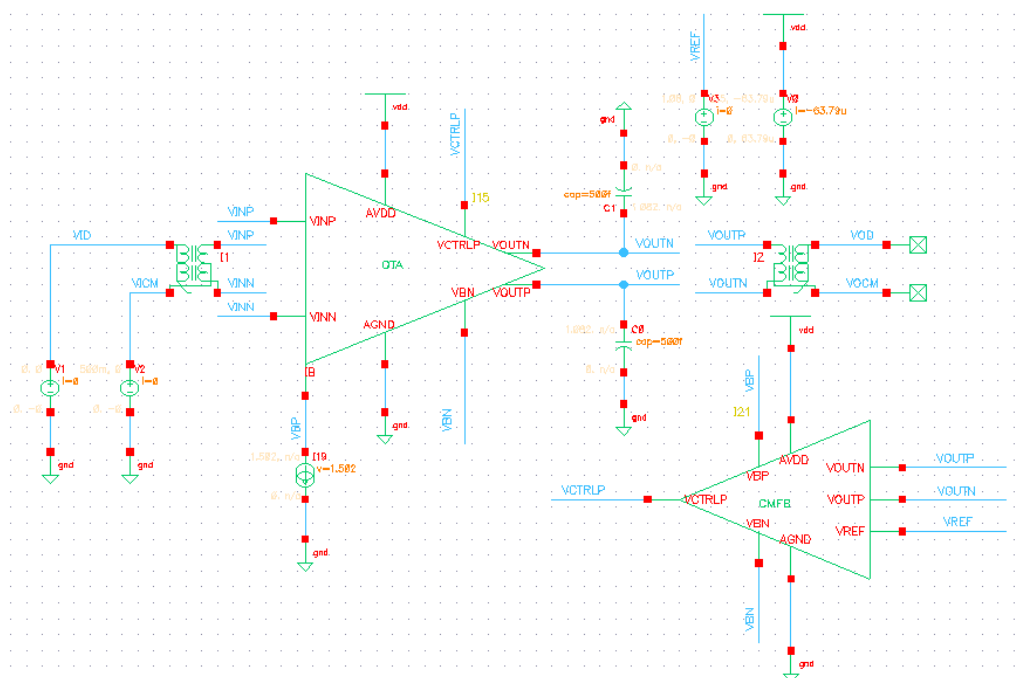


Fig.42: Biasing Voltages of Open Loop testbench

Part 4: Open-Loop OTA Simulation (Actual CMFB)

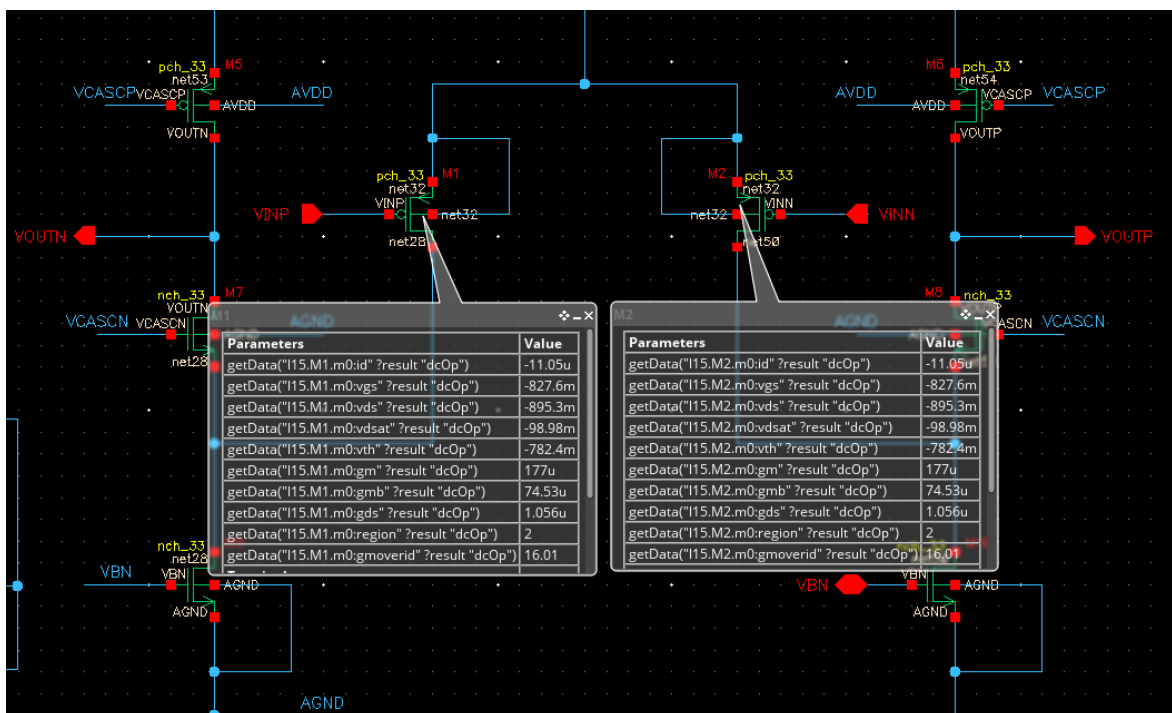


Fig.43: OP of Input pair

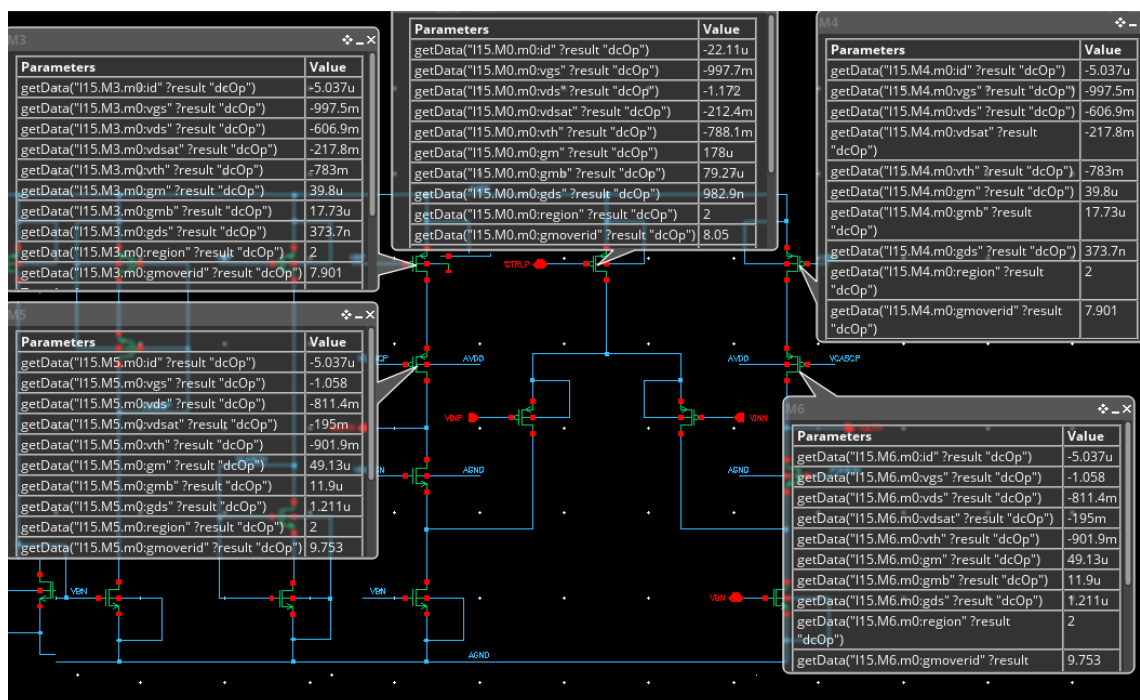


Fig.44: OP of PMOS CSs and PMOS Cascodes

Part 4: Open-Loop OTA Simulation (Actual CMFB)

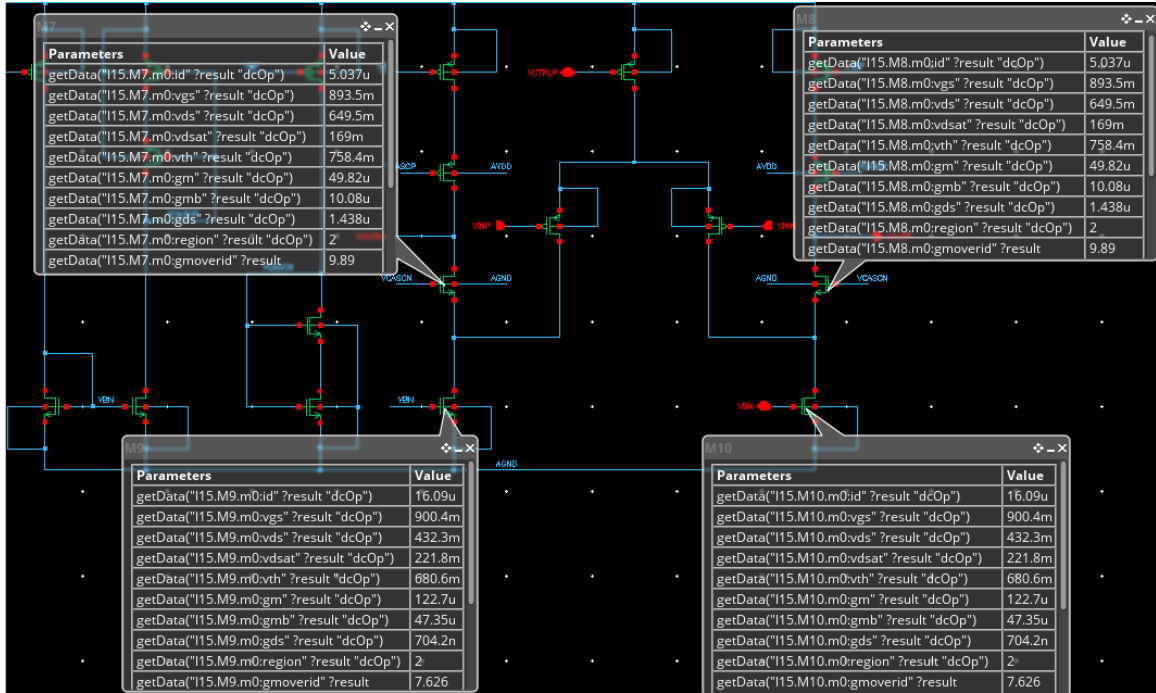


Fig.45: OP of NMOS CSs and NMOS Cascodes

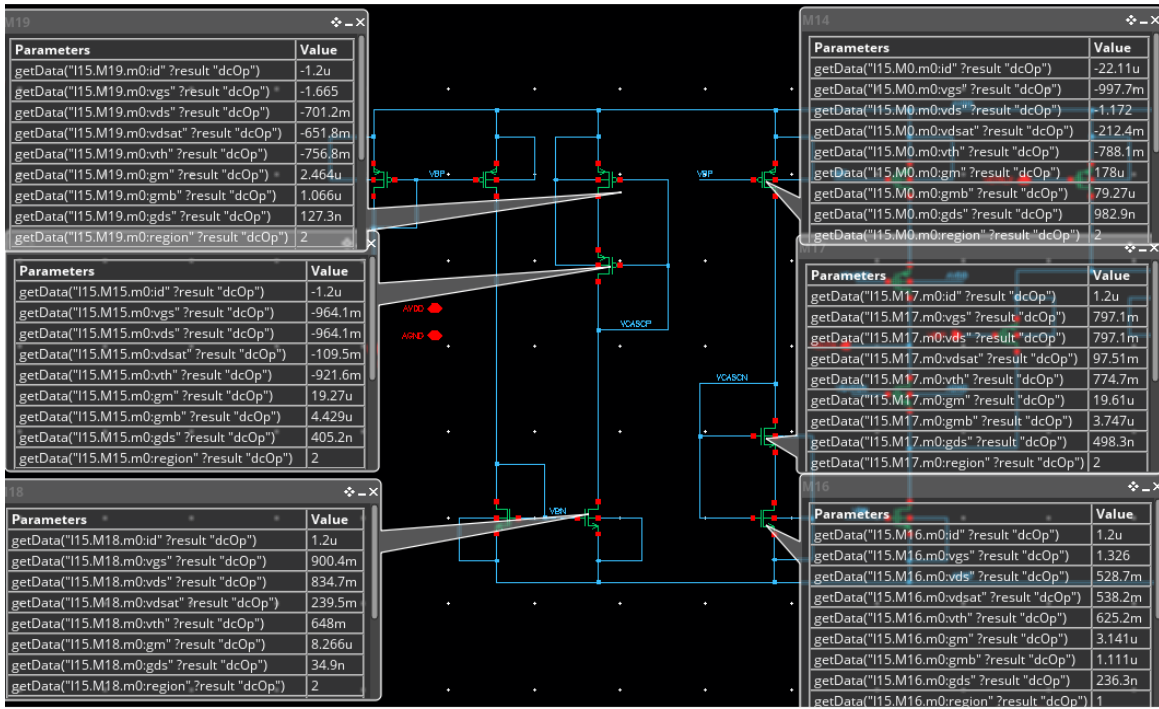


Fig.46: OP of NMOS and PMOS Magic battery

Part 4: Open-Loop OTA Simulation (Actual CMFB)

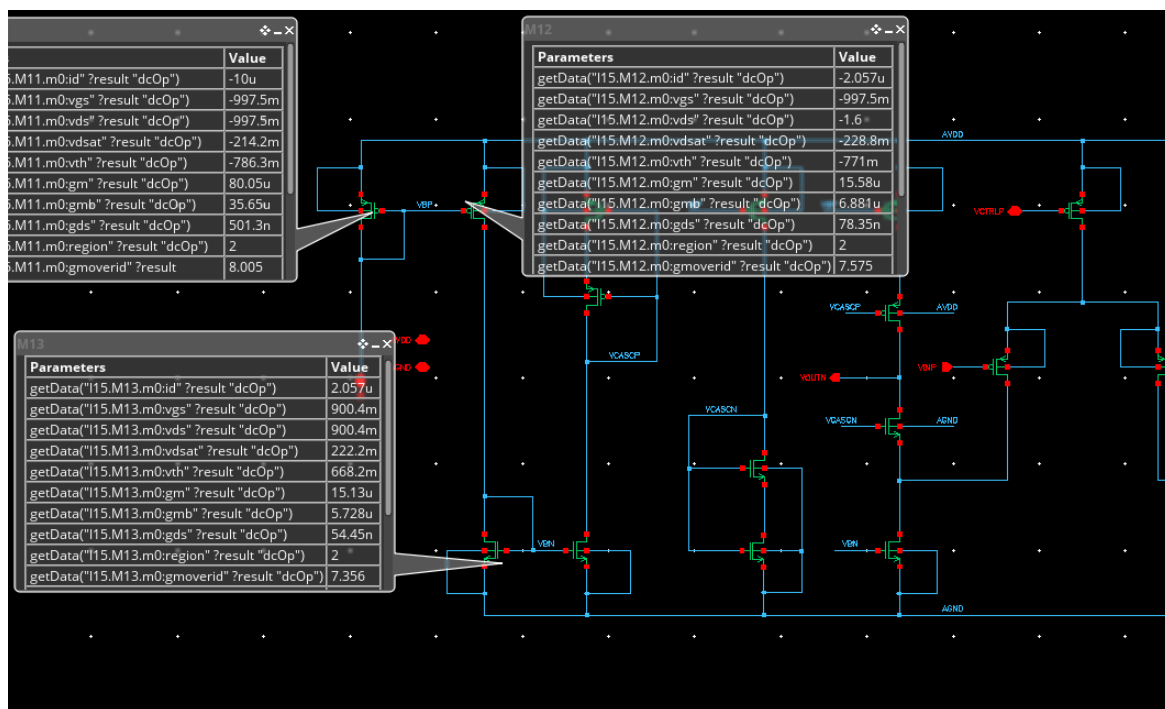


Fig.47: OP of NMOS and PMOS CM

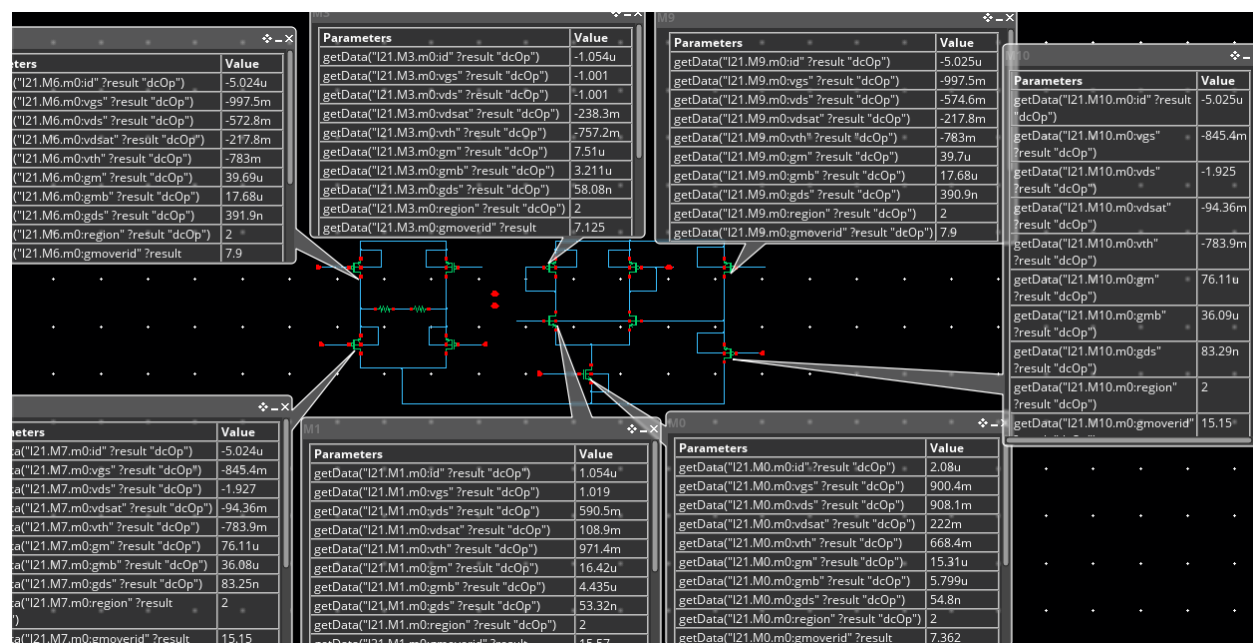


Fig.48: OP of Actual CMFB

- All Devices are in saturation except non-degenerated devices of the magic batteries.
- The Output Swing is from $0.38V$ to $1.4V$, so I choose a $V_{REF} = 1V$.

Part 4: Open-Loop OTA Simulation (Actual CMFB)

- The total Current Consumption in the Folded Cascode OTA, in current mirror branches, CMFB circuit is about $I_{B_{total}} = 63.74 \mu A$

(32.19 μA in the Folded Cascode OTA), (a 17.15 μA in the CMFB OTA)

- The dc level of common-mode output $V_{out_{CM}} = 1.002 V$, $V_{REF} = 1V$.
- What is the differential input and output voltages of the error amplifier? What is the relation between them?

$$V_{in_{diff}} = V_{GS1} - V_{GS2} = 2 mV$$

$$V_{out_{diff}} = V_{GS3} - V_{GS4} = 3.3 mV$$

$$\frac{V_{out_{diff}}}{V_{in_{diff}}} = 1.65$$

The relation between them is the error amplifier gain which is

$$A_v = \frac{g_{m1}}{g_{m3}} = 2.12$$

The A_v is not actually equal to $\left(\frac{V_{out_{diff}}}{V_{in_{diff}}}\right)$ because I neglected the effect of M_9 and M_{10}

Diff. Small Signal Ccs (AC Analysis)

Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM)

Output	Nominal	Spec	Weight	Pass/Fail
mag(v("/VOD" ?result "ac"))				
dB20(mag(v("/VOD" ?result "...				
phaseDegUnwrapped(v("/VO...				
Ao	3.487k	> 3k		pass
Ao_dB	70.85	> 70		pass
BW	15.28k	> 7.33k		pass
(Ao * BW)	53.29M	> 22M		pass
GBW	53.41M	> 22M		pass
UGF	53.15M	> 22M		pass
Cdd1+Css7+Cdd9	6.358f			
PM	79.49			
VOCM dc	1.002			
VREF dc	1			

Fig.49: diff. SS results

Part 4: Open-Loop OTA Simulation (Actual CMFB)

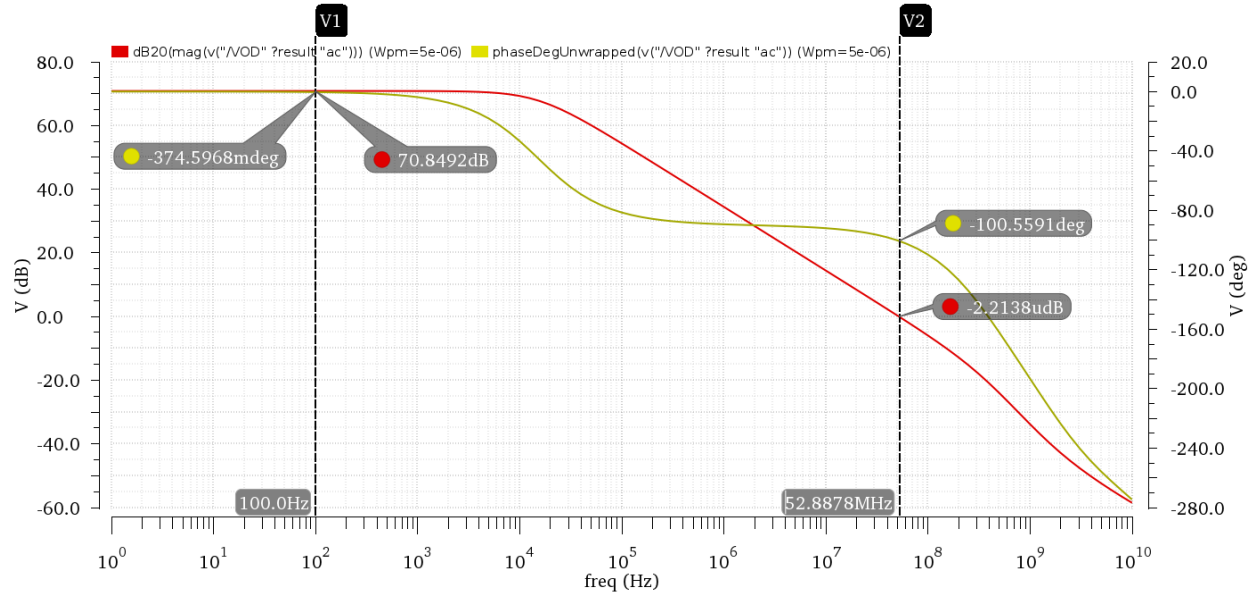


Fig.50: Bode Plot of diff. Ccs

- Hand Analysis (Note that parasitic caps are neglected cause they too small)

$$A_{vd} = G_m \times R_{out} = g_{m1,2} \times (r_{o7}(g_{m7} + g_{mb7})(r_{o9}/r_{o1})/r_{o5}(g_{m5} + g_{mb5})r_{o3})$$

$$\omega_{pd} = \frac{1}{R_{out}C_L}, \quad BW = \frac{\omega_{pd}}{2\pi}$$

$$\omega_{pnd} = \frac{g_{m7}}{C_{dd1} + C_{ss7} + C_{dd9}}$$

$$GBW = A_{vd} \times BW = \frac{g_{m1,2}}{2\pi C_L}$$

$$PM = 90^\circ - \tan^{-1}\left(\frac{\omega_u}{\omega_{pnd}}\right)$$

Parameters	Analytical	Simulation
DC Gain A_{vd}	3.566K	3.487K
DC Gain in dB A_{vd}	71.04 dB	70.85 dB
BW	15.8 KHz	15.28 KHz
GBW	56.33 MHz	53.41 MHz
UGF	56.33 MHz	53.15 MHz
Phase Margin PM	89.9°	79.49°

Table.6: diff. small signal comparison

- Hand analysis is quite different from simulation due to neglection of parasitic caps.
- the GBW in the simulation is way higher than the calculated value in the design procedure analysis which was 22 MHz. This is because we considered the feedback capacitance in the design procedure, but in this part (Open Loop) there is no feedback capacitance yet.

Part 5: Closed-Loop Simulation (AC and STB)

Create a new testbench with the OTA connected in closed-loop feedback configuration using capacitive feedback as shown below. Note that the CM DC level provided before the input (balun) is useless because it is blocked by the capacitor.

Note that we use two (baluns) at the output in order to allow stability analysis. We divide the output to diff and CM and break the diff/CM loops by 0V dc sources, then we combine them again to V_{out_n} and V_{out_p} to close the feedback loop. We use a 2pF input capacitance and 1pF feedback capacitance to provide a closed loop gain = 2. Note that you need to connect V_{out_n} to V_{in_p} and V_{out_p} to V_{in_n} to maintain negative feedback. We use large resistors across the feedback capacitance to close the loop in DC. This will set $V_{in_{CM}} = V_{out_{CM}}$.

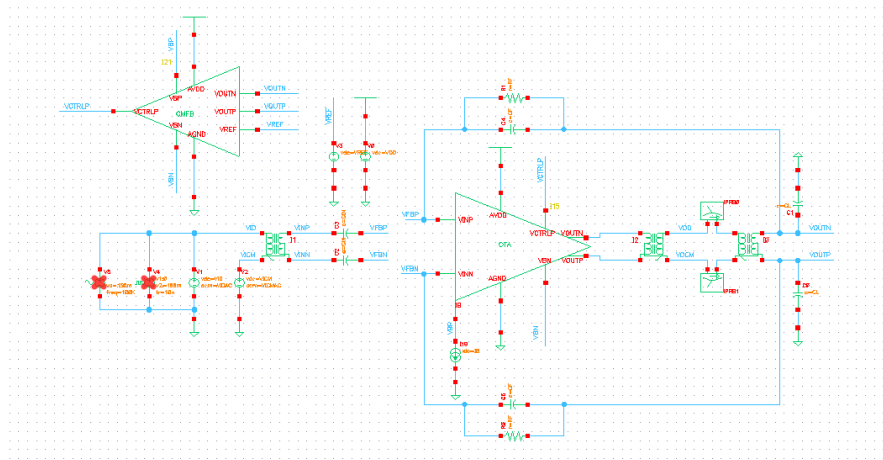


Fig.51: Closed Loop testbench

OP and Biasing Voltages

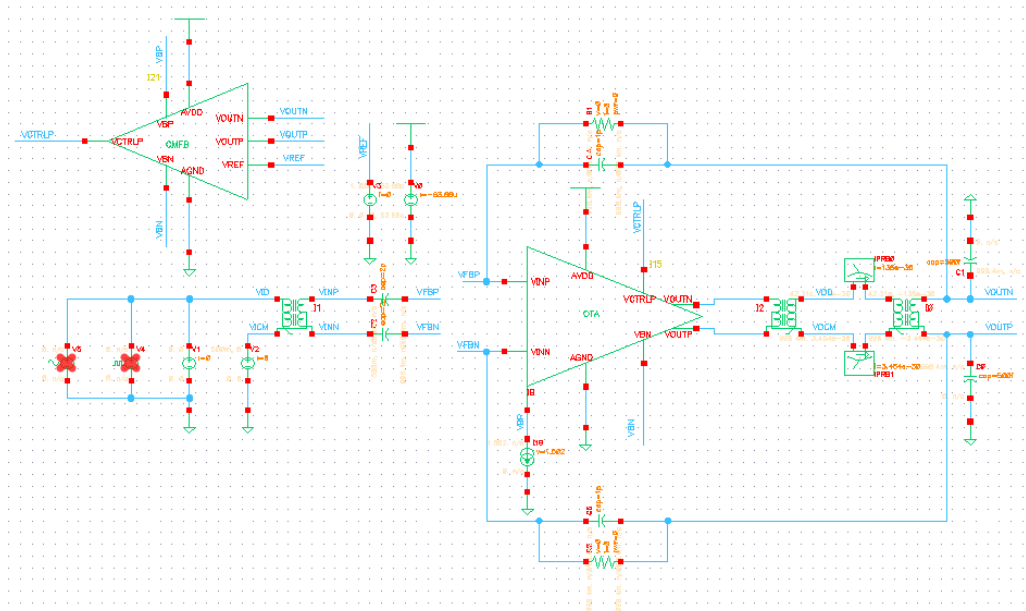


Fig.52: Biasing Voltages of Closed Loop testbench

Part 5: Closed-Loop Simulation (AC and STB)

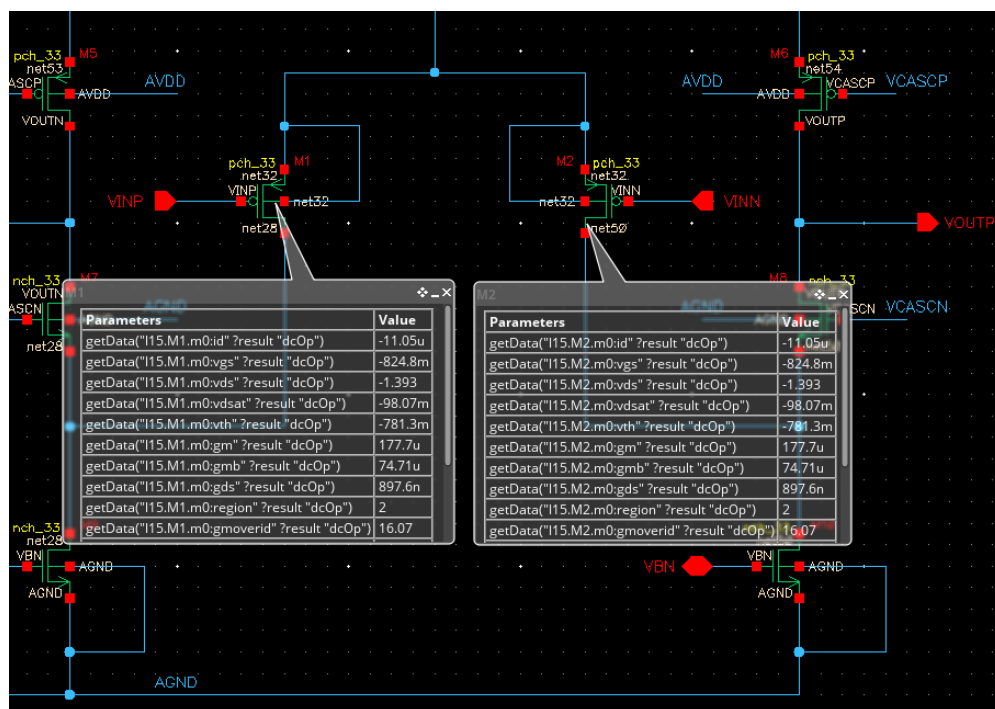


Fig.53: OP of Input pair

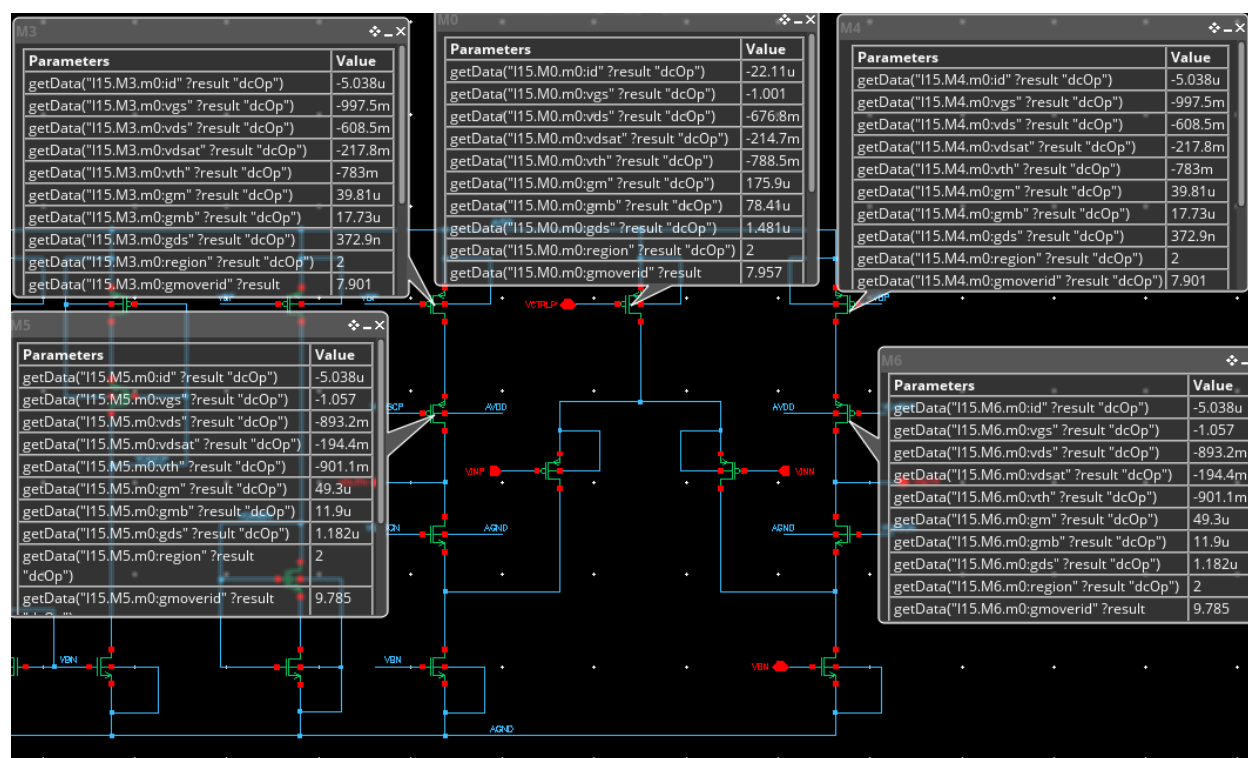


Fig.54: OP of PMOS CSs and PMOS Cascodes

Part 5: Closed-Loop Simulation (AC and STB)

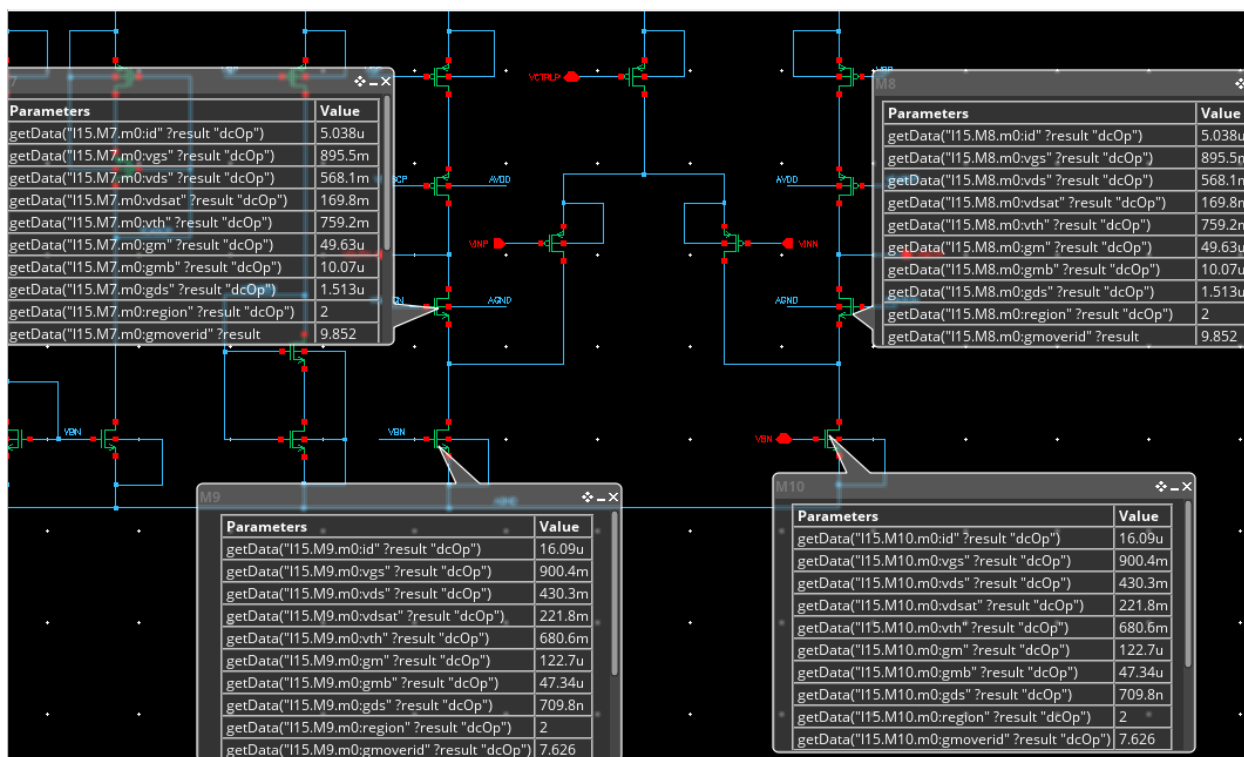


Fig.55: OP of NMOS CSs and NMOS Cascodes

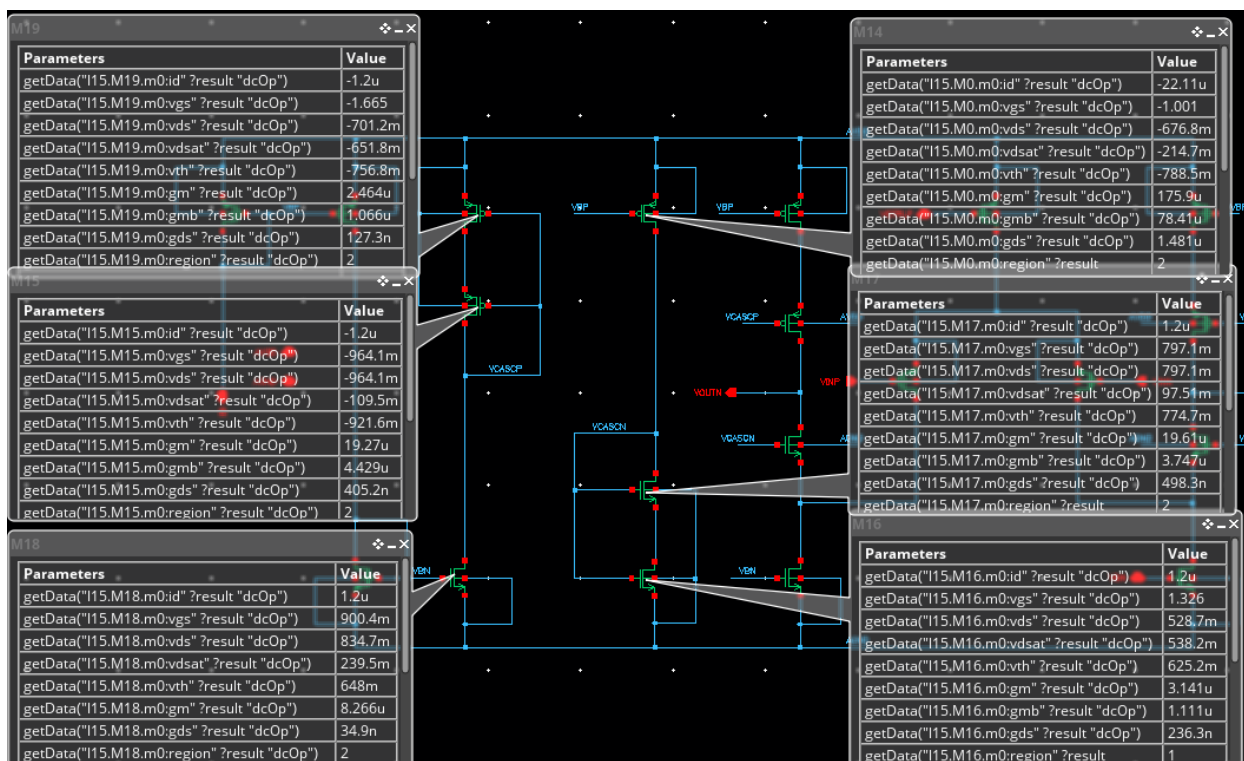


Fig.56: OP of NMOS and PMOS Magic battery

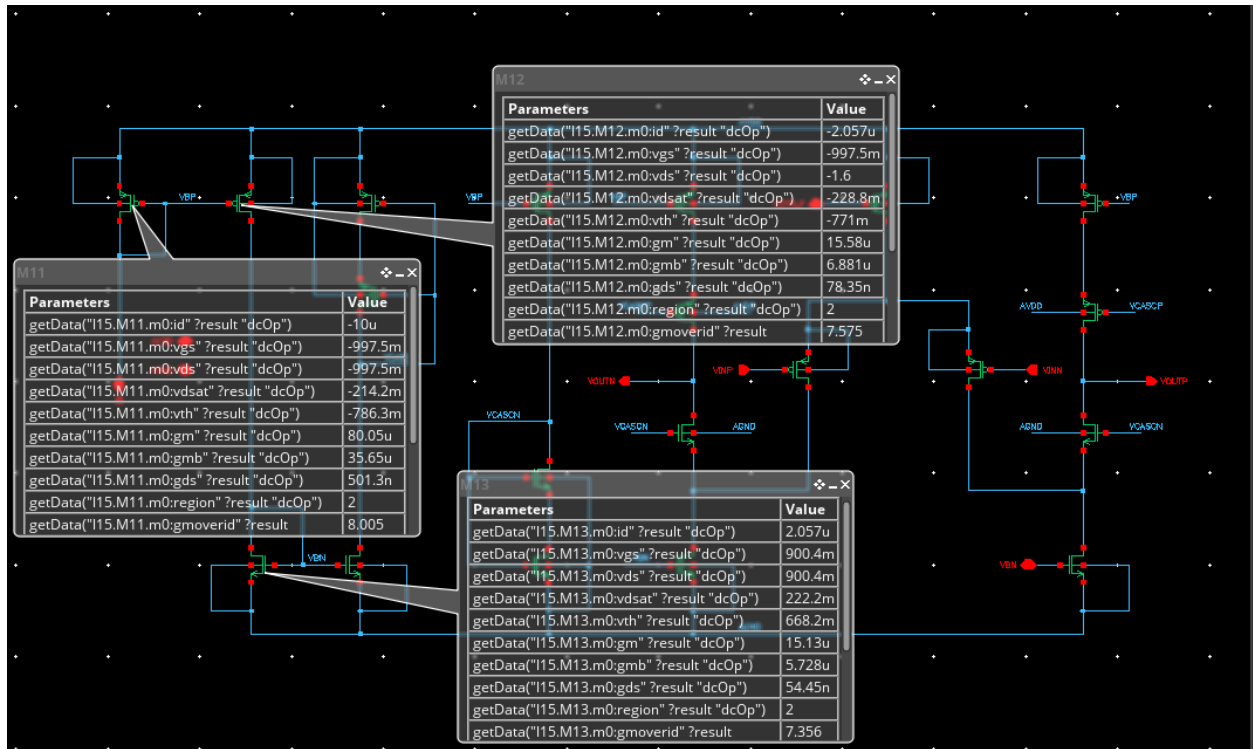


Fig.57: OP of NMOS and PMOS CM



Fig.58: OP of Actual CMFB

Part 5: Closed-Loop Simulation (AC and STB)

VOCM dc	998.4m
VREF dc	1
VICM dc	998.4m

Fig.59: dc op results

What is the CM level at the OTA Output? Why?

The CM level at the OTA output is 998.4 mV as set before close to 1V because of the finite LG of the CMFB circuit. Because the OTA amplifier works to reduce the error between the two inputs. The error is because of the finite loop gain of the CMFB circuit.

What is the CM level at the OTA Input? Why?

The CM level at the OTA input is 998.4 mV because it's in unity gain buffer configuration in DC because of the resistor shunting the output to input.

Maximum Value of V_{out} @ CMFB Circuit

$$V_{out_{max}} = V_{DD} - V_{DSAT_6} - |V_{SG_7}| = 1.4363 V$$

Minimum Value of V_{out} @ Folded OTA Circuit

$$V_{out_{min}} = V_{DSAT_7} + V_{DS_9} = 598.1mV$$

$$V_{out_{swing}} = V_{out_{max}} + V_{out_{min}} = 2.03V$$

Diff. Closed-Loop Response (AC Analysis)

Measure circuit parameters ($A_{v_{CL}}$, BW_{CL} , GBW_{CL})

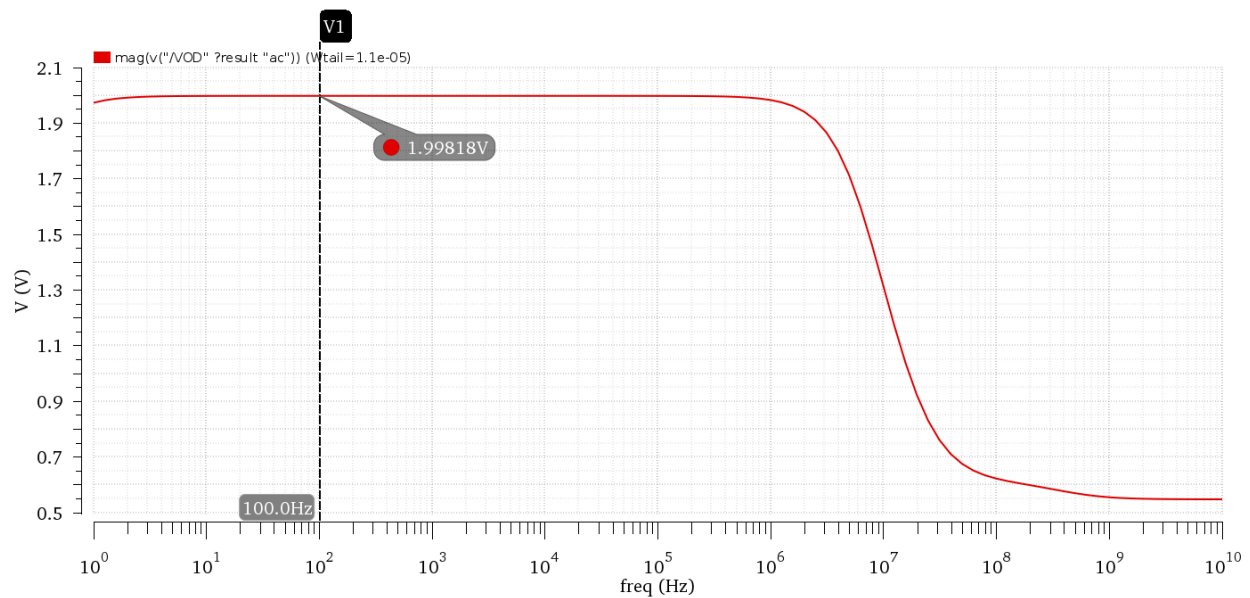


Fig.60: CL gain vs frequency

Part 5: Closed-Loop Simulation (AC and STB)

Output	Nominal	Spec	Weight	Pass/Fail
mag(v("/VOD" ?result "ac"))				
dB20(mag(v("/VOD" ?result "...				
phaseDegUnwrapped(v("/VO...				
AVCL	1.998	range 1.98 2.1		pass
AVCL dB	6.013	> 6		pass
BWCL	8.913M	> 7.33k		pass
GBWCL	17.63M	> 14.67k		pass

Fig.61: diff. CL results

Diff. and CMFB Loops Stability (STB Analysis)

Calculate circuit parameters (LG , GBW , PM)

Differential LG Path

dB20(mag(getData("loopGai...				
phaseDegUnwrapped(getDa...				
LG dB	60.91	range 60 61		pass
GBW	7.525M	> 7.33M		pass
PM	88.64	> 70		pass

Fig.62: diff. LG results

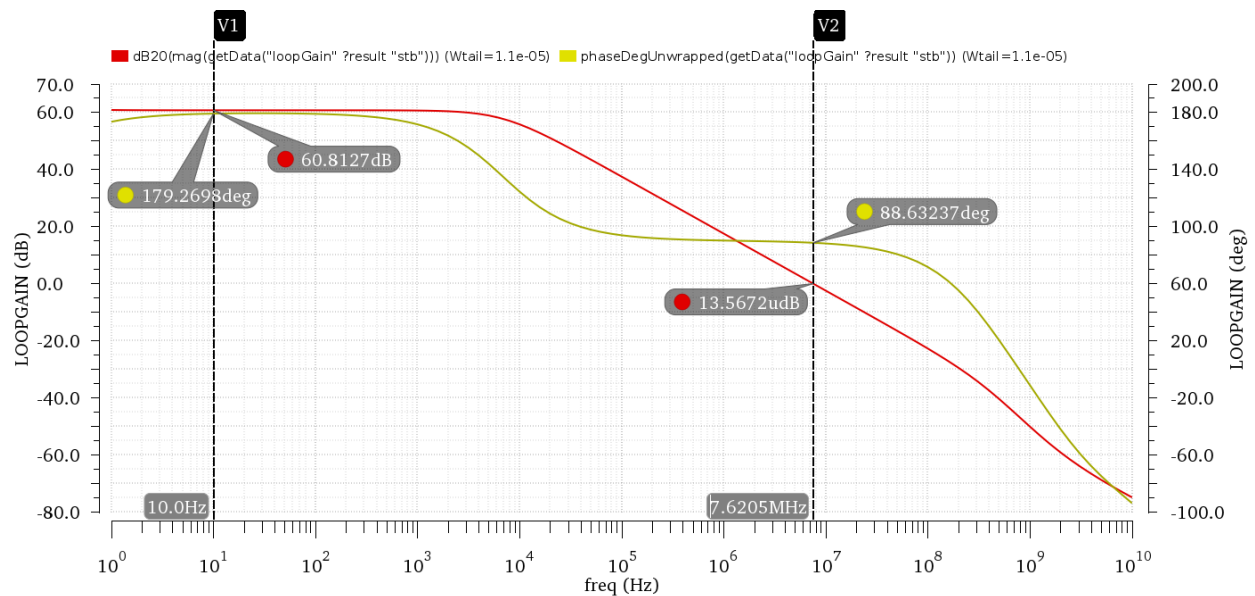


Fig.63: Bode Plot of diff. LG

CM LG Path

dB20(mag(getData("loopGai...				
phaseDegUnwrapped(getDa...				
LG dB	71.5			
GBW	12.38M	> 7.33M		pass
PM	70.53	> 70		pass

Fig.64: CM LG results

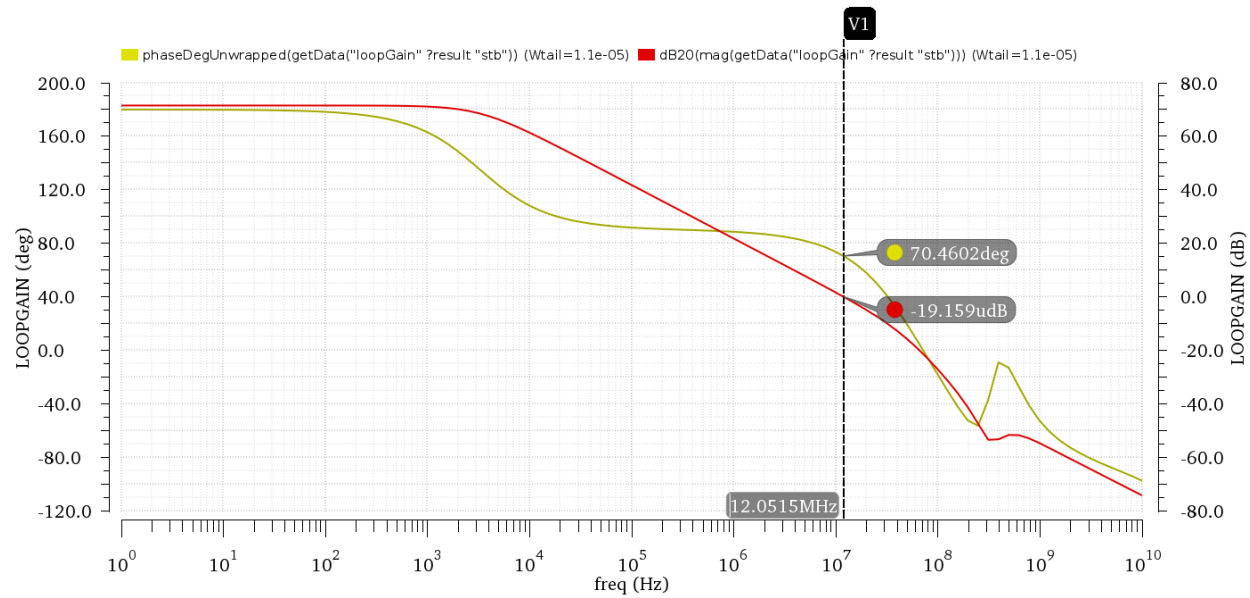


Fig.65: Bode Plot of CM LG

Part 5: Closed-Loop Simulation (AC and STB)

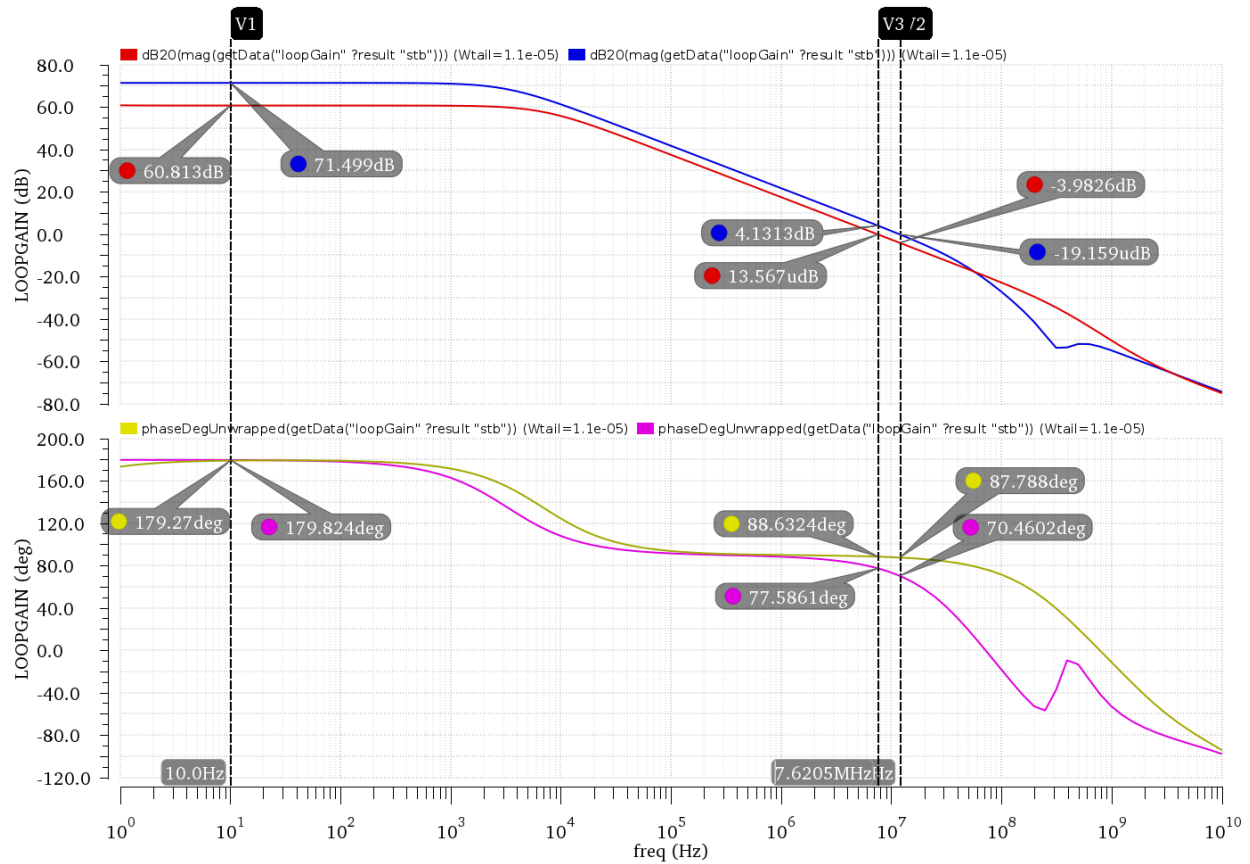


Fig.65: Bode Plot of LG

Parameters	Differential	CM
GBW	7.525 MHz	12.38 MHz
PM	88.64 °	70.53 °

Table.7: diff. and CM comparison

- The CMFB circuit has a high GBW which will make it fast enough to recover from any disturbance.
- Both diff and CM have a higher phase margin because the topology is a single gain stage having only one dominant pole acting like a first-order system.
- The CM LG is higher due to the gain of the CMFB circuit.

Parameters	Closed-Loop	Open-Loop
A_v	60.91 dB	70.85 dB
GBW	7.525 MHz	53.41 MHz

Table.8: CL and OL comparison

Part 5: Closed-Loop Simulation (AC and STB)

- The Specs from OL are mapped as following

$$LG = \beta A_{v_{OL}}$$

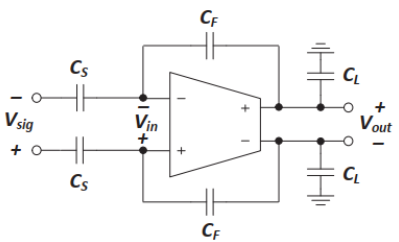
$$\beta = 0.318$$

There is an error because the feedback factor β is not exactly equal to $\frac{1}{3}$ as we drop the parasitic capacitance in our calculations

COUT	505.5a
CIN OTA	95.02f

$$\beta = \frac{C_f}{C_f + C_{in} + C_{in_{i/p\ OTA}}}$$

$$GBW_{LG} = GBW_{OL} \times \frac{C_L + C_{out}}{C_L + C_{out} + \left(C_f // (C_{in} + C_{in_{i/p\ OTA}}) \right)} \times \beta$$

$$GBW_{LG} = 8.35MHz$$


The diagram shows an OTA (Operational Transconductance Amplifier) configured with a feedback capacitor C_f between the output and the inverting input. The non-inverting input is connected to ground. The input signal V_{sig} is applied to the inverting input through a capacitor C_s . The output V_{out} is connected to ground through a capacitor C_L . The OTA is represented by a triangle with '+' and '-' signs at its inputs and output.

- GBW is different as in fact the LG is UGF not GBW so as it's not a 1st order system so they are not equal that's one of the causes.

Part 6: Closed-Loop Simulation (Transient)

Diff. and CMFB Loops (Diff. Input Pulse)

Apply a differential input pulse (initial value = 0, pulse value = 100mV, delay = 1 μ s, period = 2 μ s, pulse width = 1 μ s, rise = fall = 10 ns).

Run transient analysis for 3 μ s, with 10 ns max step.

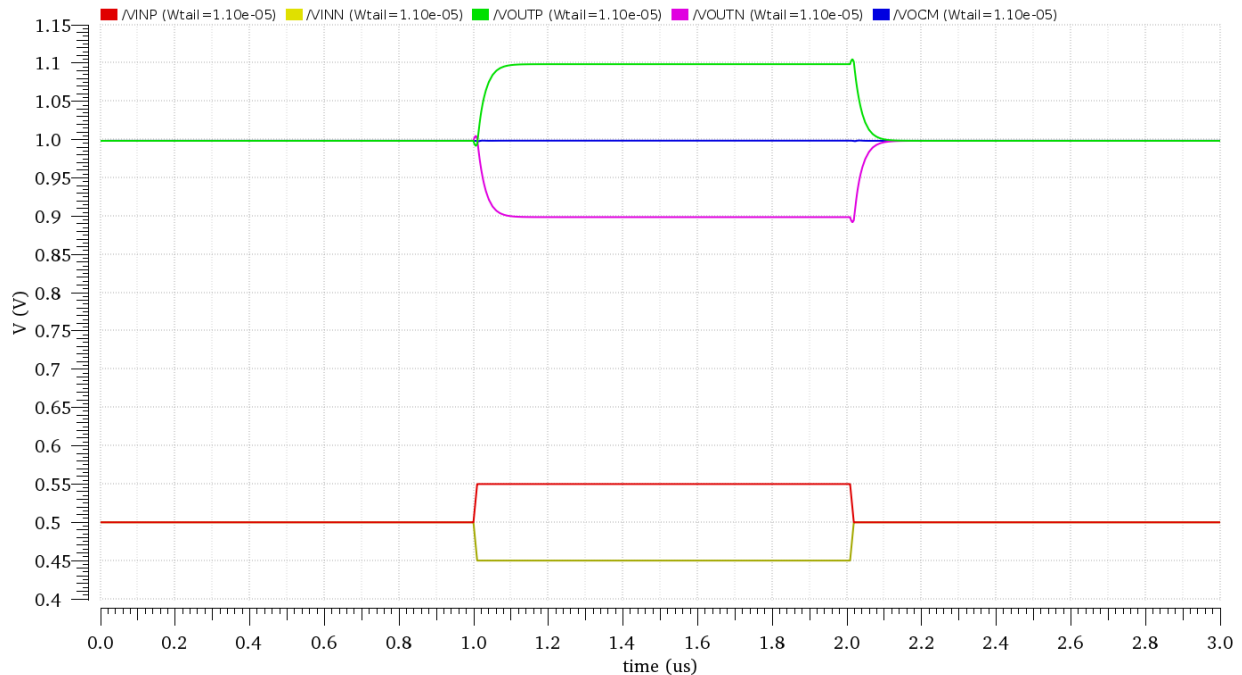


Fig.66: Diff Input and Output transient response

- Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

No, differential ringing noticed as the PM in the diff. loop is much greater than 76° and the CMFB PM is less than 76° so it will face ringing but it's very small and we can neglect it as the circuit is symmetric so the change in V_{out_n} and V_{out_p} is differentially symmetric and as the CMFB is so fast (high GBW) so its response to any change between the output nodes will be handled instantly.

- 1% Settling time

I calculated the settling time from the $A_{v_{CL}}$ value.

$$99\% \text{ of } A_{v_{CL}} = 1.972$$

Then I calculated the settling time from the graph of $V_{out_{diff}}$ vs time

Part 6: Closed-Loop Simulation (Transient)

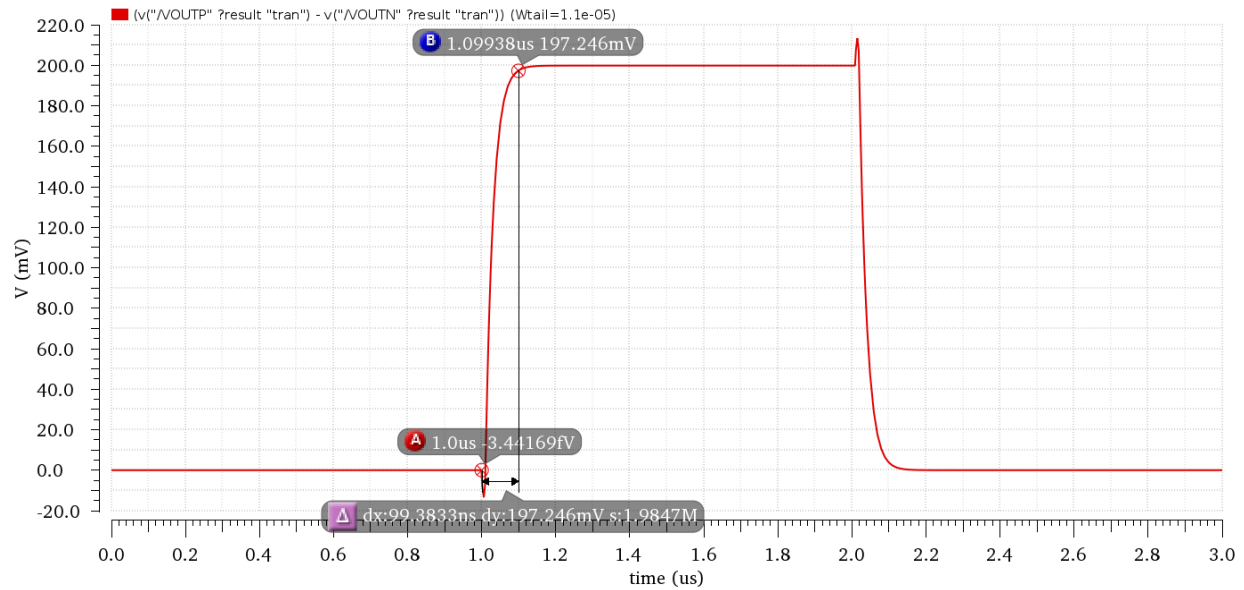


Fig.67: Diff. Output transient response

$$t_s = 99.38 \text{ ns}$$

Diff. and CMFB Loops (CM Input Pulse)

Apply a differential input pulse (initial value = 0, pulse value = 100mV, delay = 1 μ s, period = 2 μ s, pulse width = 1 μ s, rise = fall = 10 ns).

Run transient analysis for 3 μ s, with 10 ns max step.

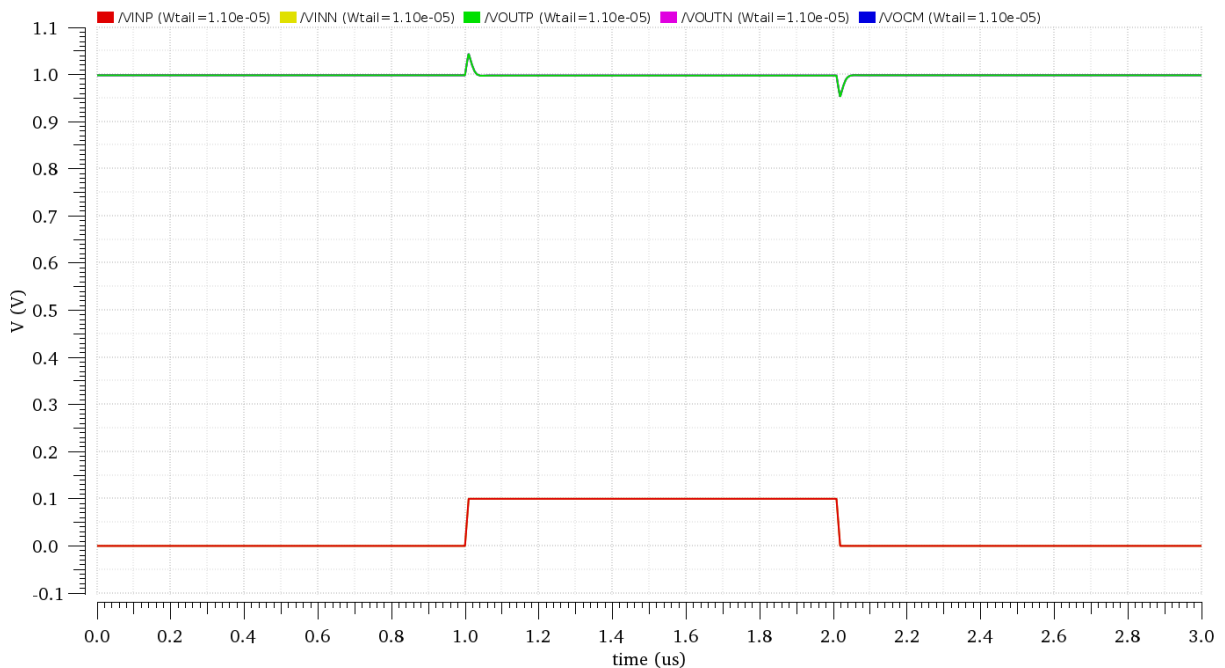


Fig.68: CM Input and Output transient response

Part 6: Closed-Loop Simulation (Transient)

- Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

No, there is no ringing in differential or CM. Yes, they both are stable with adequate PM.

- The small spark which appears in the output is due to the capacitive feedback as they make voltage divider for the first few Nano seconds in the transient response.

Output Swing

Apply a differential sinusoidal input with $freq = 100\text{ KHz}$ and $amplitude = 150\text{mV}$.

Run transient analysis for $30\text{ }\mu\text{s}$, with $(0.1\text{ }\mu\text{s})100\text{ ns}$ max step.

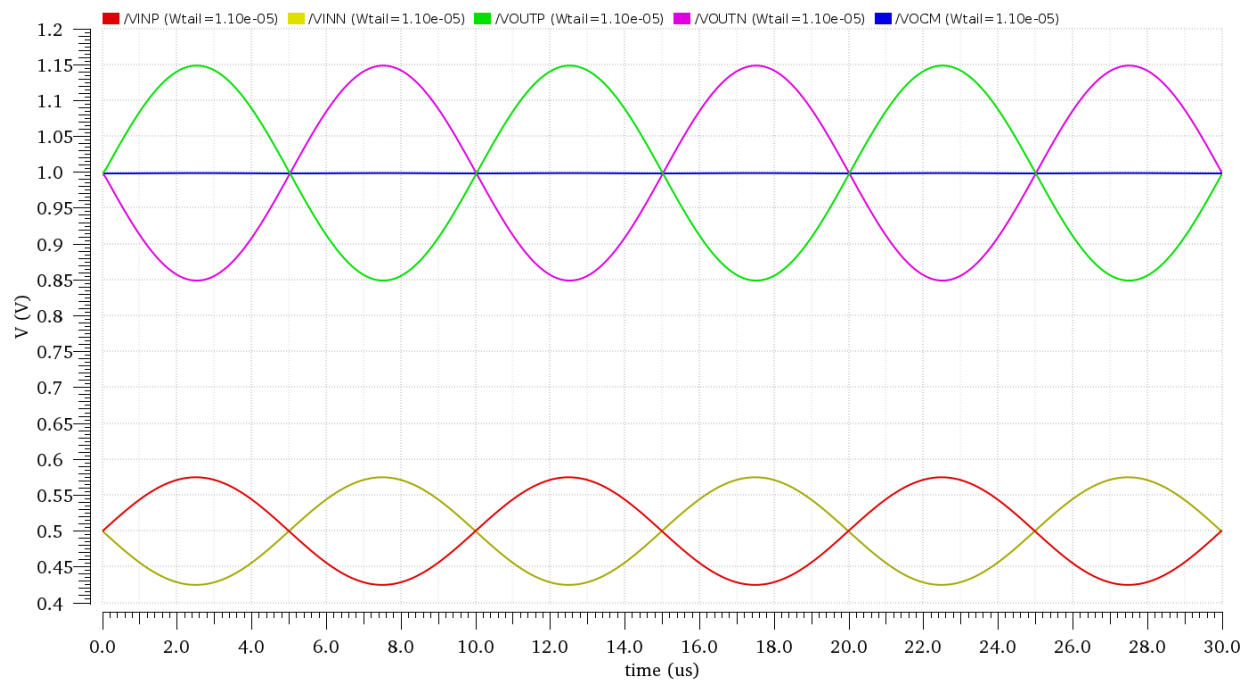


Fig.69: Input and Output transient response

Part 6: Closed-Loop Simulation (Transient)

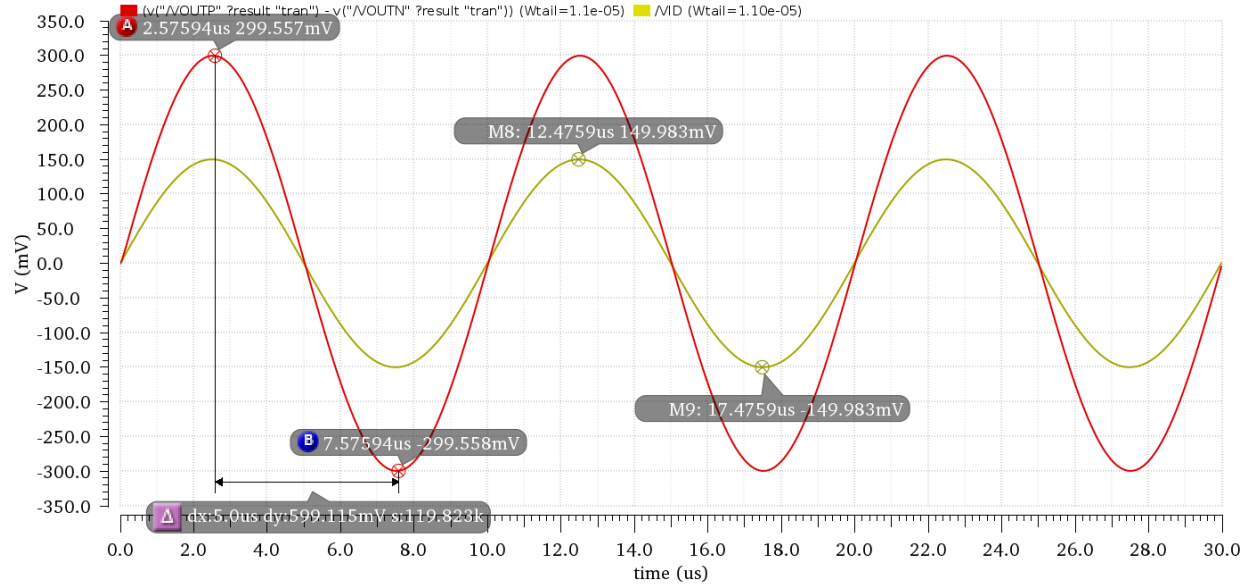


Fig.70: Diff.Input and Diff.Output transient response

(v("VOUTP" ?result "tran") - v...				
VOD P2P	599.1m			
VID P2P	300m			
AvCL	1.997			
Output Swing	1.901	> 1.2		pass

Fig.71: Transient response results

$$V_{out_{peak-to-peak}} = 599.156 \text{ mV}$$

$$V_{in_{peak-to-peak}} = 299.996 \text{ mV}$$

$$A_{v_{CL}} = \frac{V_{out_{peak-to-peak}}}{V_{in_{peak-to-peak}}} = 1.998$$

- Output Swing

$$V_{out_{swing}} = V_{DD} - V_{out_{peak-to-peak}} = 1.901 \text{ V}$$

- CMIR Range Calculations

$$V_{in_{CM_{max}}} = V_{DD} - V_{DSAT_0} - |V_{GS1,2}|$$

$$V_{in_{CM_{max}}} = 1.4605 \text{ V}$$

$$V_{in_{CM_{min}}} = -|V_{GS1,2}| + V_{DSAT_{1,2}} + V_{DSAT_{9,10}}$$

$$V_{in_{CM_{min}}} = -504.93 \text{ mV}$$

$$V_{in_{CM_{min}}} \approx -505 \text{ mV}$$

Figure of Merit

	$M_{1,2}$ Input Pair	M_0 Tail CS	$M_{3,4}$ PMOS CS	$M_{9,10}$ NMOS CS	$M_{5,6}$ NMOS Cascode	$M_{7,8}$ PMOS Cascode
W	$30 \mu m$	$11 \mu m$	$2.56 \mu m$	$5.22 \mu m$	$840 nm$	$2.16 \mu m$
L	$420 nm$	$500 nm$	$500 nm$	$1.3 \mu m$	$320 nm$	$290 nm$
Area	$25.2 (\mu m)^2$	$5.5 (\mu m)^2$	$2.56 (\mu m)^2$	$13.572 (\mu m)^2$	$0.5376 (\mu m)^2$	$1.2528 (\mu m)^2$
I_D	$11 \mu A$	$22 \mu A$	$5 \mu A$	$16 \mu A$	$5 \mu A$	$5 \mu A$

Table.9: Area and Currents of Folded Cascode OTA

	M_{11} PMOS CM	M_{12} PMOS CB	M_{13} NMOS CM	M_{14} PMOS CB	M_{18} NMOS CB	M_{16} NMOS Magic Battery	M_{19} PMOS Magic Battery
W	$5 \mu m$	$970 nm$	$711 nm$	$540 nm$	$350 nm$	$280 nm$	$270 nm$
L	$500 nm$	$500 nm$	$1.3 \mu m$	$500 nm$	$1.3 \mu m$	$6 \mu m$	$2.5 \mu m$
Area	$2.5(\mu m)^2$	$0.485(\mu m)^2$	$0.924(\mu m)^2$	$0.27(\mu m)^2$	$0.455(\mu m)^2$	$1.68(\mu m)^2$	$0.675(\mu m)^2$
I_D	$10 \mu A$	$2 \mu A$	$2 \mu A$	$1.2 \mu A$	$1.2 \mu A$	$1.2 \mu A$	$1.2 \mu A$

Table.10: Area and Currents of mirroring branches and magic battery

	$M_{1,2}$ Input Pair	M_0 Tail CS	$M_{3,4}$ PMOS CS	PMOS Input devices		$M_{5,6}$ PMOS CS	$M_{7,8}$ PMOS CS
				M_9	M_{10}		
W	$1.44 \mu m$	$711 nm$	$520 nm$	$2.56 \mu m$	$32 \mu m$	$2.56 \mu m$	$32 \mu m$
L	$1 \mu m$	$1.3 \mu m$	$500 nm$	$500 nm$	$1 \mu m$	$500 nm$	$1 \mu m$
Area	$2.88(\mu m)^2$	$0.924(\mu m)^2$	$0.52(\mu m)^2$	$1.28(\mu m)^2$	$32(\mu m)^2$	$2.56(\mu m)^2$	$64(\mu m)^2$
I_D	$1 \mu A$	$2 \mu A$	$1 \mu A$	$5 \mu A$	$5 \mu A$	$5 \mu A$	$5 \mu A$

Table.11: Area and Currents of CMFB Circuit

- Note that every two transistors have same the W/L there area is multiplied by 2 (e.g. input pair)

$$Area_{FoldedOTA} = 48.6224(\mu m)^2, I_{FoldedOTA} = 32.19 \mu A$$

$$Area_{branches} = 6.9893(\mu m)^2, I_{branches} = 14.4 \mu A$$

$$Area_{CMFB} = 104.16143(\mu m)^2, I_{CMFB} = 17.15 \mu A$$

- Figure of Merit with $GBW_{CL} = 17.63 MHz$

$$FOM = \frac{GBW_{CL}}{Area \times I_{total}} = \frac{17.63 MHz}{159.776(\mu m)^2 \times 63.74 \mu A} = 0.001731126 MHz/\mu^3 A m^2$$

Design Specs

<i>Parameter</i>	<i>Required</i>	<i>Achieved</i>
<i>Technology</i>		0.18μ
<i>Supply Voltage V_{DD}</i>		$2.5 V$
<i>Closed Loop Gain $A_{v_{CL}}$</i>	2	1.998
<i>Phase Margin (in required $A_{v_{CL}}$)</i>	$\geq 70^\circ$	88.64°
<i>CMIR – low</i>	≤ 0	$-505 mV$
<i>CMIR – high</i>	$\geq 1 V$	$1.46 V$
<i>Differential Output Swing</i>	$1.2V$ peak to peak	$1.9 V$ peak to peak
<i>DC Loop Gain LG</i>	60 dB	60.91 dB
<i>Closed Loop settling time t_s</i>	100 ns	99.38 ns
<i>Total Current Consumption I_{total}</i>	Reasonable	$63.74 \mu A$
<i>Area</i>	Reasonable	$159.77 (\mu m)^2$
$FOM = \frac{GBW_{CL}}{Area \times I_{total}}$	As High as possible	$0.001731126 MHz/\mu^3 A m^2$
<i>Chip Power Consumption</i>	Reasonable	$159.35 \mu W$
$Power = V_{DD} \times I_{total}$		

Table.12: Achieved Specs